

Part IA — Vector Calculus

Based on lectures by B. Allanach

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Lent 2015

These notes are not endorsed by the lecturers, and I have modified them (often significantly) after lectures. They are nowhere near accurate representations of what was actually lectured, and in particular, all errors are almost surely mine.

Curves in $\mathbb{R}^3$

Parameterised curves and arc length, tangents and normals to curves in $\mathbb{R}^3$, the radius of curvature. [1]

Integration in $\mathbb{R}^2$ and $\mathbb{R}^3$

Line integrals. Surface and volume integrals: definitions, examples using Cartesian, cylindrical and spherical coordinates; change of variables. [4]

Vector operators

Directional derivatives. The gradient of a real-valued function: definition; interpretation as normal to level surfaces; examples including the use of cylindrical, spherical *and general orthogonal curvilinear* coordinates.

Divergence, curl and ∇^2 in Cartesian coordinates, examples; formulae for these operators (statement only) in cylindrical, spherical *and general orthogonal curvilinear* coordinates. Solenoidal fields, irrotational fields and conservative fields; scalar potentials. Vector derivative identities. [5]

Integration theorems

Divergence theorem, Green's theorem, Stokes's theorem, Green's second theorem: statements; informal proofs; examples; application to fluid dynamics, and to electromagnetism including statement of Maxwell's equations. [5]

Laplace's equation

Laplace's equation in $\mathbb{R}^2$ and $\mathbb{R}^3$: uniqueness theorem and maximum principle. Solution of Poisson's equation by Gauss's method (for spherical and cylindrical symmetry) and as an integral. [4]

Cartesian tensors in $\mathbb{R}^3$

Tensor transformation laws, addition, multiplication, contraction, with emphasis on tensors of second rank. Isotropic second and third rank tensors. Symmetric and antisymmetric tensors. Revision of principal axes and diagonalization. Quotient theorem. Examples including inertia and conductivity. [5]

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0 Introduction

In the differential equations class, we learnt how to do calculus in one dimension. However, (apparently) the world has more than one dimension. We live in a 3 (or 4) dimensional world, and string theorists think that the world has more than 10 dimensions. It is thus important to know how to do calculus in many dimensions.

For example, the position of a particle in a three dimensional world can be given by a position vector $\mathbf{x}$. Then by definition, the velocity is given by $\frac{d}{dt}\mathbf{x} = \dot{\mathbf{x}}$. This would require us to take the derivative of a vector.

This is not too difficult. We can just differentiate the vector componentwise. However, we can reverse the problem and get a more complicated one. We can assign a number to each point in (3D) space, and ask how this number changes as we move in space. For example, the function might tell us the temperature at each point in space, and we want to know how the temperature changes with position.

In the most general case, we will assign a vector to each point in space. For example, the electric field vector $\mathbf{E}(\mathbf{x})$ tells us the direction of the electric field at each point in space.

On the other side of the story, we also want to do integration in multiple dimensions. Apart from the obvious “integrating a vector”, we might want to integrate over surfaces. For example, we can let $\mathbf{v}(\mathbf{x})$ be the velocity of some fluid at each point in space. Then to find the total fluid flow through a surface, we integrate $\mathbf{v}$ over the surface.

In this course, we are mostly going to learn about doing calculus in many dimensions. In the last few lectures, we are going to learn about Cartesian tensors, which is a generalization of vectors.

Note that throughout the course (and lecture notes), summation convention is implied unless otherwise stated.

1 Derivatives and coordinates

1.1 Derivative of functions

Remark (Motivation). In single-variable calculus, the derivative of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ at a point $x \in \mathbb{R}$ is defined as the limit

$$f'(x) = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x},$$

provided the limit exists. This definition cannot be directly generalized to vector-valued functions, since division by a vector is not defined. We therefore seek an alternative characterization of differentiability that extends naturally to higher dimensions.

Recall that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at x , then for a small perturbation $\delta x \in \mathbb{R}$, we have

$$\delta f \stackrel{\text{def}}{=} f(x + \delta x) - f(x) = f'(x)\delta x + o(\delta x),$$

where $o(\delta x)$ denotes a quantity such that $o(\delta x)/\delta x \rightarrow 0$ as $\delta x \rightarrow 0$. This says that the change in f is, to first order, a linear function of the perturbation δx . Conversely, if f satisfies this relation for some constant $f'(x)$, then f is differentiable at x .

This characterization extends naturally to functions involving vectors. In the most general case, where both the domain and codomain are vector spaces, the perturbation $\delta \mathbf{x}$ is a vector, and the “proportionality constant” relating $\delta \mathbf{F}$ to $\delta \mathbf{x}$ is a matrix. This matrix is what we call the derivative.

Vector functions $\mathbb{R} \rightarrow \mathbb{R}^n$

Definition (Vector function). A *vector function* is a function $\mathbf{F} : \mathbb{R} \rightarrow \mathbb{R}^n$. It takes in a scalar and returns a vector. For example, it can map a time $t \in \mathbb{R}$ to the velocity of a particle at that time.

Definition (Derivative of vector function). A vector function $\mathbf{F} : \mathbb{R} \rightarrow \mathbb{R}^n$ is *differentiable* at $x \in \mathbb{R}$ if there exists a vector $\mathbf{F}'(x) \in \mathbb{R}^n$ such that

$$\delta \mathbf{F} \stackrel{\text{def}}{=} \mathbf{F}(x + \delta x) - \mathbf{F}(x) = \mathbf{F}'(x)\delta x + o(\delta x).$$

The vector $\mathbf{F}'(x)$ is called the *derivative* of $\mathbf{F}$ at x .

Remark. Since δx is a scalar, dividing by it is well-defined. The derivative can therefore equivalently be expressed as

$$\mathbf{F}'(x) = \frac{d\mathbf{F}}{dx} = \lim_{\delta x \rightarrow 0} \frac{1}{\delta x} [\mathbf{F}(x + \delta x) - \mathbf{F}(x)].$$

In differential notation, the differentiability condition becomes $d\mathbf{F} =$

$$\mathbf{F}'(x) \, dx.$$

Proposition (Componentwise derivative). Let $\mathbf{F} : \mathbb{R} \rightarrow \mathbb{R}^n$ be a differentiable vector function, and let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be a basis for $\mathbb{R}^n$ that is independent of x . Writing $\mathbf{F}(x) = F_i(x)\mathbf{e}_i$, the derivative is given componentwise by

$$\mathbf{F}'(x) = F'_i(x)\mathbf{e}_i.$$

Proof. Writing $\mathbf{F}(x) = F_i(x)\mathbf{e}_i$ and using the definition of the derivative,

$$\mathbf{F}'(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbf{F}(x + \delta x) - \mathbf{F}(x)}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{F_i(x + \delta x) - F_i(x)}{\delta x} \mathbf{e}_i = F'_i(x) \mathbf{e}_i,$$

where we used that the basis vectors $\mathbf{e}_i$ are independent of x . $\square$

Proposition (Leibniz rules for products). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable scalar function, and let $\mathbf{g}, \mathbf{h} : \mathbb{R} \rightarrow \mathbb{R}^3$ be differentiable vector functions. Then the following product rules hold:

$$\begin{aligned} \frac{d}{dt}(f\mathbf{g}) &= \frac{df}{dt}\mathbf{g} + f\frac{d\mathbf{g}}{dt}, \\ \frac{d}{dt}(\mathbf{g} \cdot \mathbf{h}) &= \frac{d\mathbf{g}}{dt} \cdot \mathbf{h} + \mathbf{g} \cdot \frac{d\mathbf{h}}{dt}, \\ \frac{d}{dt}(\mathbf{g} \times \mathbf{h}) &= \frac{d\mathbf{g}}{dt} \times \mathbf{h} + \mathbf{g} \times \frac{d\mathbf{h}}{dt}. \end{aligned}$$

In the case of the cross product, the order of multiplication must be retained.

Proof. Each identity follows from expanding the relevant product componentwise and applying the ordinary product rule for scalar functions. For example, for the dot product, writing $\mathbf{g} = g_i\mathbf{e}_i$ and $\mathbf{h} = h_i\mathbf{e}_i$ gives $\mathbf{g} \cdot \mathbf{h} = g_i h_i$, and so

$$\frac{d}{dt}(\mathbf{g} \cdot \mathbf{h}) = \frac{d}{dt}(g_i h_i) = g'_i h_i + g_i h'_i = \frac{d\mathbf{g}}{dt} \cdot \mathbf{h} + \mathbf{g} \cdot \frac{d\mathbf{h}}{dt}.$$

The other two identities are proved similarly. $\square$

Example (Particle momentum and torque). Consider a particle of mass m with position $\mathbf{r}(t)$, velocity $\dot{\mathbf{r}}(t)$, and acceleration $\ddot{\mathbf{r}}(t)$, where dots denote derivatives with respect to time t . Its momentum is $\mathbf{p} = m\dot{\mathbf{r}}(t)$. If $\mathbf{F}(\mathbf{r})$ is the force on the particle, then Newton's second law states that

$$\dot{\mathbf{p}} = m\ddot{\mathbf{r}} = \mathbf{F}.$$

The *angular momentum* about the origin is defined by

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = m\mathbf{r} \times \dot{\mathbf{r}}.$$

Differentiating and applying the Leibniz rule for the cross product gives

$$\dot{\mathbf{L}} = m\dot{\mathbf{r}} \times \dot{\mathbf{r}} + m\mathbf{r} \times \ddot{\mathbf{r}} = m\mathbf{r} \times \ddot{\mathbf{r}} = \mathbf{r} \times \mathbf{F},$$

where we used $\dot{\mathbf{r}} \times \dot{\mathbf{r}} = \mathbf{0}$. The quantity $\mathbf{r} \times \mathbf{F}$ is called the *torque* of $\mathbf{F}$ about the origin.

Scalar functions $\mathbb{R}^n \rightarrow \mathbb{R}$

Definition (Scalar function). A *scalar function* is a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. It takes in a position vector and returns a scalar. For example, a scalar function may assign to each point in space the temperature or the potential energy at that point.

Definition (Limits and asymptotic notation). Let $\mathbf{v}$ and $\mathbf{c}$ be vectors in $\mathbb{R}^n$. We write $\mathbf{v} \rightarrow \mathbf{c}$ if $|\mathbf{v} - \mathbf{c}| \rightarrow 0$. For a function g (scalar- or vector-valued) of a vector variable $\mathbf{h} \in \mathbb{R}^n$, we write $g(\mathbf{h}) = o(\mathbf{h})$ to mean $|g(\mathbf{h})|/|\mathbf{h}| \rightarrow 0$ as $\mathbf{h} \rightarrow \mathbf{0}$.

Definition (Gradient of scalar function). A scalar function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *differentiable* at $\mathbf{r} \in \mathbb{R}^n$ if there exists a vector $\nabla f \in \mathbb{R}^n$ such that

$$\delta f \stackrel{\text{def}}{=} f(\mathbf{r} + \delta \mathbf{r}) - f(\mathbf{r}) = (\nabla f) \cdot \delta \mathbf{r} + o(\delta \mathbf{r})$$

for all perturbations $\delta \mathbf{r} \in \mathbb{R}^n$. The vector ∇f is called the *gradient* of f at $\mathbf{r}$.

Remark. Unlike the case of vector functions $\mathbb{R} \rightarrow \mathbb{R}^n$, the gradient cannot be defined simply by dividing δf by $\delta \mathbf{r}$, since division by a vector is not defined. The definition above avoids this by characterizing the derivative through its linear action on perturbations.

Definition (Directional derivative). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable scalar function, let $\mathbf{r} \in \mathbb{R}^n$, and let $\mathbf{n}$ be a unit vector in $\mathbb{R}^n$. The *directional derivative* of f in the direction $\mathbf{n}$ at the point $\mathbf{r}$ is

$$D_{\mathbf{n}}f(\mathbf{r}) = \mathbf{n} \cdot \nabla f = \lim_{h \rightarrow 0} \frac{1}{h} [f(\mathbf{r} + h\mathbf{n}) - f(\mathbf{r})].$$

It measures the rate of change of f with respect to displacement in the direction $\mathbf{n}$.

Remark. The directional derivative is obtained by restricting the perturbation in the definition of differentiability to the form $\delta \mathbf{r} = h\mathbf{n}$, where $h \in \mathbb{R}$ is a small scalar. Substituting into the definition gives

$$f(\mathbf{r} + h\mathbf{n}) - f(\mathbf{r}) = \nabla f \cdot (h\mathbf{n}) + o(h) = h(\nabla f \cdot \mathbf{n}) + o(h),$$

from which the formula $D_{\mathbf{n}}f = \mathbf{n} \cdot \nabla f$ follows upon dividing by h and

taking $h \rightarrow 0$.

Remark (Geometric interpretation of the gradient). Since $\mathbf{n} \cdot \nabla f = |\nabla f| \cos \theta$, where θ is the angle between $\mathbf{n}$ and ∇f , the directional derivative is maximized when $\mathbf{n}$ points in the same direction as ∇f , giving $D_{\mathbf{n}}f = |\nabla f|$. Thus, ∇f points in the direction of steepest increase of f , and $|\nabla f|$ is the rate of increase in that direction.

Theorem (Gradient in Cartesian coordinates). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable scalar function, and let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be an orthonormal basis for $\mathbb{R}^n$ with corresponding Cartesian coordinates $(x_1, \dots, x_n)$. Then the gradient is given by

$$\nabla f = \frac{\partial f}{\partial x_i} \mathbf{e}_i = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \mathbf{e}_i.$$

Proof. Setting $\mathbf{n} = \mathbf{e}_i$ in the definition of the directional derivative gives

$$\mathbf{e}_i \cdot \nabla f = D_{\mathbf{e}_i} f = \lim_{h \rightarrow 0} \frac{1}{h} [f(\mathbf{r} + h\mathbf{e}_i) - f(\mathbf{r})] = \frac{\partial f}{\partial x_i}.$$

Since $\{\mathbf{e}_i\}$ is an orthonormal basis, we can expand $\nabla f = (\mathbf{e}_i \cdot \nabla f) \mathbf{e}_i$ (with summation over i), which gives the result. $\square$

Remark (Differential notation). Using the gradient formula, the differentiability condition can be written as

$$\delta f = \frac{\partial f}{\partial x_i} \delta x_i + o(\delta \mathbf{r}).$$

In differential notation, this becomes

$$df = \nabla f \cdot d\mathbf{r} = \frac{\partial f}{\partial x_i} dx_i,$$

which expresses the chain rule for partial derivatives.

Example (Computing a gradient). Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined by $f(x, y, z) = x + e^{xy} \sin z$. Then

$$\begin{aligned} \nabla f &= \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \\ &= (1 + ye^{xy} \sin z, xe^{xy} \sin z, e^{xy} \cos z). \end{aligned}$$

At the point $(x, y, z) = (0, 1, 0)$, we find $\nabla f = (1, 0, 1)$. The function f increases most rapidly in the direction $\mathbf{n} = \frac{1}{\sqrt{2}}(1, 0, 1)$, with a rate of change of $|\nabla f| = \sqrt{2}$. Conversely, f decreases most rapidly in the direction $-\mathbf{n}$. There is no change in f for directions perpendicular to ∇f .

Theorem (Chain rule for scalar functions). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable scalar function and let $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^n$ be a differentiable path parametrized by u . Then the composition $f(\mathbf{r}(u))$ is differentiable as a function of u , and

$$\frac{df}{du} = \nabla f \cdot \frac{d\mathbf{r}}{du} = \frac{\partial f}{\partial x_i} \frac{dx_i}{du}.$$

Proof. A small change δu produces a change $\delta \mathbf{r} = \mathbf{r}'(u)\delta u + o(\delta u)$ in the position. By the differentiability of f ,

$$\delta f = \nabla f \cdot \delta \mathbf{r} + o(|\delta \mathbf{r}|) = \nabla f \cdot \mathbf{r}'(u)\delta u + o(\delta u).$$

Dividing by δu and taking $\delta u \rightarrow 0$ gives the result. $\square$

Remark. In differential notation, the chain rule can be written as $df = \nabla f \cdot d\mathbf{r} = \frac{\partial f}{\partial x_i} dx_i$, recovering the formula from the earlier remark.

Vector fields $\mathbb{R}^n \rightarrow \mathbb{R}^m$

Definition (Vector field). A *vector field* is a function $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$. It assigns a vector in $\mathbb{R}^m$ to each point of $\mathbb{R}^n$.

Definition (Derivative of vector field). A vector field $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *differentiable* at $\mathbf{x} \in \mathbb{R}^n$ if there exists an $m \times n$ matrix M such that

$$\delta \mathbf{F} \stackrel{\text{def}}{=} \mathbf{F}(\mathbf{x} + \delta \mathbf{x}) - \mathbf{F}(\mathbf{x}) = M\delta \mathbf{x} + o(\delta \mathbf{x}).$$

The matrix M is called the *derivative* of $\mathbf{F}$ at $\mathbf{x}$.

Theorem (Jacobian matrix). Let $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a differentiable vector field with component functions $F_j : \mathbb{R}^n \rightarrow \mathbb{R}$ for $j = 1, \dots, m$, so that $\mathbf{F}(\mathbf{x}) = (F_1(\mathbf{x}), \dots, F_m(\mathbf{x}))$. Writing $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = \mathbf{F}(\mathbf{x}) = (y_1, \dots, y_m)$, the entries of the derivative matrix M are given by

$$M_{ji} = \frac{\partial y_j}{\partial x_i} = \frac{\partial F_j}{\partial x_i}.$$

The matrix M is called the *Jacobian matrix* of $\mathbf{F}$.

Proof. Each component $y_j = F_j(\mathbf{x})$ is a scalar function of $\mathbf{x}$. By the gradient formula,

$$dy_j = \frac{\partial F_j}{\partial x_i} dx_i.$$

Writing this in matrix form as $d\mathbf{y} = M d\mathbf{x}$, we read off $M_{ji} = \frac{\partial F_j}{\partial x_i}$. $\square$

Remark. While the Jacobian formula could serve as an alternative definition of the derivative, the coordinate-free definition given above is preferable because it does not depend on the choice of coordinate system.

Definition (Smooth function). A function is *smooth* if all partial derivatives of all orders exist and are continuous.

Proposition (Schwarz's theorem). If f is smooth, then its mixed partial derivatives are symmetric:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

for all i, j . That is, the order of differentiation does not matter.

Remark. In this course, all functions will be assumed smooth except where they are obviously singular (e.g. $f(x) = 1/x$ at $x = 0$).

Theorem (Chain rule). Let $g : \mathbb{R}^p \rightarrow \mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable functions, and let the coordinates of $\mathbb{R}^p$, $\mathbb{R}^n$, and $\mathbb{R}^m$ be denoted by u_a , x_i , and y_r respectively. Then the composition $f \circ g : \mathbb{R}^p \rightarrow \mathbb{R}^m$ is differentiable, and

$$\frac{\partial y_r}{\partial u_a} = \frac{\partial y_r}{\partial x_i} \frac{\partial x_i}{\partial u_a},$$

with summation over i implied. In matrix form,

$$M(f \circ g)_{ra} = M(f)_{ri} M(g)_{ia}.$$

Alternatively, in operator form,

$$\frac{\partial}{\partial u_a} = \frac{\partial x_i}{\partial u_a} \frac{\partial}{\partial x_i}.$$

Proof. A small perturbation $\delta \mathbf{u}$ produces a change $\delta \mathbf{x} = M(g) \delta \mathbf{u} + o(\delta \mathbf{u})$ by the differentiability of g . By the differentiability of f ,

$$\delta \mathbf{y} = M(f) \delta \mathbf{x} + o(\delta \mathbf{x}) = M(f) M(g) \delta \mathbf{u} + o(\delta \mathbf{u}),$$

since $|\delta \mathbf{x}|/|\delta \mathbf{u}|$ remains bounded as $\delta \mathbf{u} \rightarrow \mathbf{0}$. This shows $M(f \circ g) = M(f) M(g)$, which gives the component formula $\frac{\partial y_r}{\partial u_a} = \frac{\partial y_r}{\partial x_i} \frac{\partial x_i}{\partial u_a}$. $\square$

1.2 Inverse functions

Proposition (Derivative of inverse function). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be differentiable functions satisfying $g \circ f = f \circ g = \text{id}$ (so that $g = f^{-1}$). If $f(\mathbf{x}) = \mathbf{u}$ and $g(\mathbf{u}) = \mathbf{x}$, then

$$M(g) = M(f)^{-1},$$

where $M(f)$ and $M(g)$ denote the Jacobian matrices of f and g , respectively.

Proof. The derivative of the identity function $\text{id} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the identity matrix I , since $\frac{\partial u_b}{\partial u_a} = \delta_{ab}$. By the chain rule applied to $g \circ f = \text{id}$, we have

$$M(g \circ f) = M(g)M(f) = I.$$

Therefore $M(g) = M(f)^{-1}$.

In terms of partial derivatives, the chain rule gives

$$\frac{\partial u_b}{\partial x_i} \frac{\partial x_i}{\partial u_a} = \delta_{ab},$$

which is the component form of the matrix equation $M(f)M(g) = I$. □

Remark. In the one-dimensional case $n = 1$, this reduces to the familiar formula $\frac{du}{dx} = 1/\frac{dx}{du}$.

Example (Polar coordinates Jacobian). Consider the case $n = 2$. Define coordinates $u_1 = \rho$, $u_2 = \varphi$ and Cartesian coordinates x_1, x_2 related by $x_1 = \rho \cos \varphi$ and $x_2 = \rho \sin \varphi$. The coordinate transformation is then given by the function $g(\rho, \varphi) = (\rho \cos \varphi, \rho \sin \varphi)$, with Jacobian matrix

$$M(g) = \begin{pmatrix} \partial x_1 / \partial \rho & \partial x_1 / \partial \varphi \\ \partial x_2 / \partial \rho & \partial x_2 / \partial \varphi \end{pmatrix} = \begin{pmatrix} \cos \varphi & -\rho \sin \varphi \\ \sin \varphi & \rho \cos \varphi \end{pmatrix}.$$

Inverting the relations gives

$$\rho = \sqrt{x_1^2 + x_2^2}, \quad \varphi = \tan^{-1} \frac{x_2}{x_1}.$$

The Jacobian matrix of the inverse function $f = g^{-1}$ is then

$$M(f) = \begin{pmatrix} \partial \rho / \partial x_1 & \partial \rho / \partial x_2 \\ \partial \varphi / \partial x_1 & \partial \varphi / \partial x_2 \end{pmatrix} = M(g)^{-1},$$

as guaranteed by the proposition above.

Definition (Jacobian). The *Jacobian* of a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the determinant of its Jacobian matrix, $\det M(f)$.

Corollary (Jacobian product formula). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $g = f^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be differentiable inverse functions. Then

$$\det M(f) \cdot \det M(g) = 1.$$

That is, the Jacobians of inverse functions are reciprocals of each other.

Proof. Taking determinants of $M(g)M(f) = I$ and using $\det(AB) = \det A \cdot \det B$ gives $\det M(g) \cdot \det M(f) = 1$. $\square$

1.3 Coordinate systems

Definition (Curvilinear coordinates). Let $(x_1, \dots, x_n)$ denote Cartesian coordinates in $\mathbb{R}^n$ with respect to an orthonormal basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$. A *curvilinear coordinate system* $(u_1, \dots, u_n)$ is defined by specifying smooth, invertible functions relating the two sets of coordinates.

Example (Plane polar coordinates). In two dimensions, *plane polar coordinates* (ρ, φ) are defined by

$$x_1 = \rho \cos \varphi, \quad x_2 = \rho \sin \varphi,$$

where $\rho \geq 0$ is the radial distance from the origin and $\varphi \in [0, 2\pi)$ is the angle measured from the positive x_1 -axis.

Definition (Basis vectors for curvilinear coordinates). Given curvilinear coordinates $(u_1, \dots, u_n)$, the associated *basis vectors* are defined by differentiating the position vector with respect to each coordinate and normalizing:

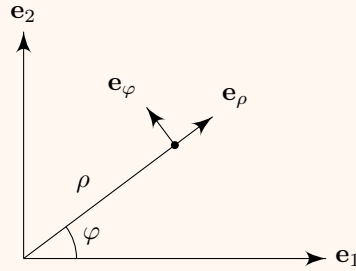
$$\mathbf{e}_{u_i} = \frac{\partial \mathbf{r} / \partial u_i}{|\partial \mathbf{r} / \partial u_i|}.$$

These basis vectors point in the direction of increasing u_i at each point.

Remark. The curvilinear coordinates u_i are not in general the components of the position vector with respect to the basis $\{\mathbf{e}_{u_i}\}$. For instance, in polar coordinates, $\mathbf{r} \neq \rho \mathbf{e}_\rho + \varphi \mathbf{e}_\varphi$, even though $\mathbf{r} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2$ in Cartesian coordinates.

Example (Polar coordinate basis vectors). For plane polar coordinates (ρ, φ) , the basis vectors are

$$\mathbf{e}_\rho = \cos \varphi \mathbf{e}_1 + \sin \varphi \mathbf{e}_2, \quad \mathbf{e}_\varphi = -\sin \varphi \mathbf{e}_1 + \cos \varphi \mathbf{e}_2.$$



Remark. Unlike the Cartesian basis vectors $\mathbf{e}_1, \mathbf{e}_2$, the curvilinear basis vectors $\mathbf{e}_{u_i}$ vary with position and may be undefined at certain points (e.g., $\mathbf{e}_\rho$ and $\mathbf{e}_\varphi$ are undefined at the origin). Despite this, they are extremely useful for problems with appropriate symmetry.

Example (Cylindrical and spherical polar coordinates). In three dimensions, the most commonly used curvilinear coordinate systems are *cylindrical polar coordinates* (ρ, φ, z) and *spherical polar coordinates* (r, θ, φ) . Their conversion formulae and associated basis vectors are as follows.

Cylindrical polars	Spherical polars
Conversion formulae	
$x_1 = \rho \cos \varphi$	$x_1 = r \sin \theta \cos \varphi$
$x_2 = \rho \sin \varphi$	$x_2 = r \sin \theta \sin \varphi$
$x_3 = z$	$x_3 = r \cos \theta$
Basis vectors	
$\mathbf{e}_\rho = (\cos \varphi, \sin \varphi, 0)$	$\mathbf{e}_r = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$
$\mathbf{e}_\varphi = (-\sin \varphi, \cos \varphi, 0)$	$\mathbf{e}_\varphi = (-\sin \varphi, \cos \varphi, 0)$
$\mathbf{e}_z = (0, 0, 1)$	$\mathbf{e}_\theta = (\cos \theta \cos \varphi, \cos \theta \sin \varphi, -\sin \theta)$

2 Curves and Line

2.1 Parametrised curves, lengths and arc length

Remark (Motivation). A curve in $\mathbb{R}^n$ can be described in several ways. One approach is to specify an equation that points on the curve satisfy; for example, a circle can be described by $x^2 + y^2 = 1$. However, this implicit description is often difficult to work with computationally, and for many curves no simple closed-form equation exists.

A more practical approach is to describe a curve parametrically, as the trajectory of a point moving through space. This parametric description has the additional benefit of assigning an *orientation* to the curve, corresponding to the direction of increasing parameter.

Definition (Parametrisation of curve). Let C be a curve in $\mathbb{R}^n$. A *parametrisation* of C is a continuous, piecewise differentiable, and invertible function $\mathbf{r} : D \rightarrow \mathbb{R}^n$, where $D \subseteq \mathbb{R}$ is an interval, such that the image of $\mathbf{r}$ is C .

At points where $\mathbf{r}$ is differentiable, the derivative $\mathbf{r}'(u)$ is a vector tangent to the curve. A parametrisation is called *regular* if $\mathbf{r}'(u) \neq \mathbf{0}$ for all $u \in D$ where the derivative exists.

Remark. A given curve admits infinitely many parametrisations. For instance, if $\mathbf{r}(u)$ parametrises C , then so does $\mathbf{r}(\phi(v))$ for any smooth, invertible function $\phi : D' \rightarrow D$.

Example (Ellipse parametrisation). Consider the upper half of an ellipse lying in the plane $z = 3$:

$$\frac{x^2}{4} + y^2 = 1, \quad y \geq 0, \quad z = 3.$$

This curve can be parametrised by $\mathbf{r}(u) = 2 \cos u \hat{\mathbf{i}} + \sin u \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ for $u \in [0, \pi]$.

Proposition (Arc length derivative). Let $\mathbf{r}(u)$ be a regular parametrisation of a curve C , and let $s(u)$ denote the arc length measured along C from some fixed reference point. Then

$$\frac{ds}{du} = \pm \left| \frac{d\mathbf{r}}{du} \right| = \pm |\mathbf{r}'(u)|,$$

where the sign is positive if s increases with u (i.e., arc length is measured in the direction of the parametrisation) and negative otherwise.

Proof. A small change δu in the parameter produces a change $\delta \mathbf{r} = \mathbf{r}'(u)\delta u + o(\delta u)$

in the position. The corresponding change in arc length is $\delta s = |\delta \mathbf{r}| = |\mathbf{r}'(u)| |\delta u| + o(\delta u)$. Dividing by δu and taking the limit gives the result, with the sign determined by whether s and u increase in the same direction. $\square$

Example (Helix arclength). Consider a helix in $\mathbb{R}^3$ described by $\mathbf{r}(u) = (3 \cos u, 3 \sin u, 4u)$ for $u \geq 0$. The tangent vector is

$$\mathbf{r}'(u) = (-3 \sin u, 3 \cos u, 4),$$

and its magnitude is

$$|\mathbf{r}'(u)| = \sqrt{9 \sin^2 u + 9 \cos^2 u + 16} = \sqrt{9 + 16} = 5.$$

Thus $\frac{ds}{du} = 5$, and the arc length from $\mathbf{r}(0)$ to $\mathbf{r}(u)$ is $s = 5u$.

Remark (Reparametrisation). Given a parametrisation $\mathbf{r}(u)$ of a curve C , we can obtain a new parametrisation by composing with any smooth, invertible function $\tilde{u} = \phi(u)$. The new parametrisation is $\tilde{\mathbf{r}}(\tilde{u}) = \mathbf{r}(\phi^{-1}(\tilde{u}))$, and by the chain rule,

$$\frac{d\mathbf{r}}{d\tilde{u}} = \frac{d\mathbf{r}}{du} \cdot \frac{du}{d\tilde{u}}.$$

Corollary (Arc-length parametrisation). If a curve C is parametrised by its arc length s , then the tangent vector has unit length:

$$|\mathbf{r}'(s)| = \left| \frac{d\mathbf{r}}{ds} \right| = 1.$$

Proof. Setting $u = s$ in the arclength derivative formula gives $\frac{ds}{ds} = 1 = |\mathbf{r}'(s)|$. $\square$

Definition (Scalar line element). The *scalar line element* of a curve C is the infinitesimal arc length ds . For a curve parametrised by $\mathbf{r}(u)$, we have

$$ds = |\mathbf{r}'(u)| du.$$

2.2 Line integrals of vector fields

Definition (Line integral). Let $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a continuous vector field and let C be a curve parametrised by $\mathbf{r} : [\alpha, \beta] \rightarrow \mathbb{R}^n$. The *line integral* of $\mathbf{F}$ along C in the direction from $\mathbf{r}(\alpha)$ to $\mathbf{r}(\beta)$ is

$$\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_{\alpha}^{\beta} \mathbf{F}(\mathbf{r}(u)) \cdot \mathbf{r}'(u) du.$$

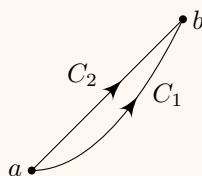
The quantity $d\mathbf{r} = \mathbf{r}'(u) du$ is called the *line element* (or *vector line*

element) on C .

Remark (Physical interpretation). Consider a particle moving from point $\mathbf{a}$ to point $\mathbf{b}$ along a curve C under the influence of a force field $\mathbf{F}$. The work done by the force is the sum of contributions $\mathbf{F}(\mathbf{r}) \cdot \delta \mathbf{r}$ over infinitesimal displacements $\delta \mathbf{r}$ along the curve. In the limit, the total work done is

$$W = \int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r}.$$

Example (Path dependence of line integrals). Let $\mathbf{F}(\mathbf{r}) = (xe^y, z^2, xy)$ and consider the line integral of $\mathbf{F}$ from $\mathbf{a} = (0, 0, 0)$ to $\mathbf{b} = (1, 1, 1)$ along two different paths.



Path C_1 : Parametrise by $\mathbf{r}(u) = (u, u^2, u^3)$ for $u \in [0, 1]$. Then $\mathbf{r}'(u) = (1, 2u, 3u^2)$ and $\mathbf{F}(\mathbf{r}(u)) = (ue^{u^2}, u^6, u^3)$, so

$$\begin{aligned} \int_{C_1} \mathbf{F} \cdot d\mathbf{r} &= \int_0^1 \mathbf{F} \cdot \mathbf{r}'(u) \, du = \int_0^1 ue^{u^2} + 2u^7 + 3u^5 \, du \\ &= \frac{e}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{2} = \frac{e}{2} + \frac{1}{4}. \end{aligned}$$

Path C_2 : Parametrise by $\mathbf{r}(t) = (t, t, t)$ for $t \in [0, 1]$. Then $\mathbf{r}'(t) = (1, 1, 1)$ and $\mathbf{F}(\mathbf{r}(t)) = (te^t, t^2, t^2)$, so

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_0^1 te^t + 2t^2 \, dt = \frac{5}{3}.$$

Since the two integrals differ, the line integral of $\mathbf{F}$ depends on the path, not just the endpoints $\mathbf{a}$ and $\mathbf{b}$.

Remark (Arclength parametrisation). If we use the arc length s as the parameter, then $d\mathbf{r} = \mathbf{t} \, ds$, where $\mathbf{t} = \mathbf{r}'(s)$ is the unit tangent vector. The line integral becomes

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{t} \, ds.$$

Definition (Scalar line integral). For a scalar function f and a curve C , the *scalar line integral* of f along C with respect to arc length is

$$\int_C f \, ds.$$

By convention, this integral is evaluated in the direction of increasing arc length.

Remark. As a special case, the length of the curve C is given by $\int_C 1 \, ds$.

Definition (Closed curve and circulation). A *closed curve* is a curve whose start and end points coincide. The line integral of a vector field $\mathbf{F}$ around a closed curve C is often written as $\oint_C \mathbf{F} \cdot d\mathbf{r}$ and is called the *circulation* of $\mathbf{F}$ around C .

Definition (Piecewise smooth curve). A *piecewise smooth curve* is a curve $C = C_1 + C_2 + \cdots + C_n$ where each C_i is smooth with a regular parametrisation. The line integral over a piecewise smooth curve C is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} + \cdots + \int_{C_n} \mathbf{F} \cdot d\mathbf{r}.$$

If C is a curve traversed from $\mathbf{a}$ to $\mathbf{b}$, we write $-C$ for the same curve traversed in the opposite direction, from $\mathbf{b}$ to $\mathbf{a}$.

Proposition (Reversal of line integral). If C is a curve from $\mathbf{a}$ to $\mathbf{b}$, then

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = - \int_C \mathbf{F} \cdot d\mathbf{r}.$$

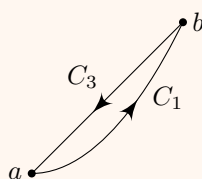
Proof. If $\mathbf{r}(u)$ for $u \in [\alpha, \beta]$ parametrises C , then $\tilde{\mathbf{r}}(u) = \mathbf{r}(\alpha + \beta - u)$ parametrises $-C$. Since $\tilde{\mathbf{r}}'(u) = -\mathbf{r}'(\alpha + \beta - u)$, the substitution $v = \alpha + \beta - u$ gives

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = \int_{\alpha}^{\beta} \mathbf{F}(\tilde{\mathbf{r}}(u)) \cdot \tilde{\mathbf{r}}'(u) \, du = - \int_{\alpha}^{\beta} \mathbf{F}(\mathbf{r}(v)) \cdot \mathbf{r}'(v) \, dv = - \int_C \mathbf{F} \cdot d\mathbf{r}. \quad \square$$

Example (Piecewise smooth closed curve). Continuing the previous example, let $C_3 = -C_2$ denote the straight-line path from $\mathbf{b}$ to $\mathbf{a}$. Then

$C = C_1 + C_3$ is a piecewise smooth closed curve, and

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_3} \mathbf{F} \cdot d\mathbf{r} \\ &= \left(\frac{e}{2} + \frac{1}{4} \right) - \frac{5}{3} \\ &= -\frac{17}{12} + \frac{e}{2}.\end{aligned}$$



2.3 Gradients and Differentials

As the previous example demonstrates, line integrals generally depend on the path taken, not just the endpoints. However, there is an important class of vector fields for which the line integral depends only on the endpoints. These are the vector fields that can be expressed as gradients of scalar functions.

Theorem (Fundamental theorem for line integrals). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable scalar function and let C be a piecewise smooth curve from $\mathbf{a}$ to $\mathbf{b}$. If $\mathbf{F} = \nabla f$, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a}).$$

In particular, the line integral depends only on the endpoints and not on the path. As a special case, if C is a closed curve, then $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$.

Remark. This result is the multivariable analogue of the fundamental theorem of calculus.

Proof. Let $\mathbf{r}(u)$ be any parametrization of the curve, and suppose $\mathbf{a} = \mathbf{r}(\alpha)$, $\mathbf{b} = \mathbf{r}(\beta)$. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = \int_{\alpha}^{\beta} \nabla f \cdot \frac{d\mathbf{r}}{du} du.$$

So by the chain rule, this is equal to

$$\int_{\alpha}^{\beta} \frac{d}{du}(f(\mathbf{r}(u))) du = [f(\mathbf{r}(u))]_{\alpha}^{\beta} = f(\mathbf{b}) - f(\mathbf{a}). \quad \square$$

Definition (Conservative vector field). A vector field $\mathbf{F}$ is called *conservative* if $\mathbf{F} = \nabla f$ for some scalar function f . The function f is called a *potential* for $\mathbf{F}$.

Remark (Origin of the term “conservative”). The terminology comes from mechanics: if a force field $\mathbf{F}$ is conservative, then the work done around any closed loop is zero, which implies that energy is conserved. A particle cannot gain net energy by travelling around a closed path.

It is often useful to work directly with the differential form $\mathbf{F} \cdot d\mathbf{r} = F_i dx_i$, treating it as an object that can be integrated along curves.

Definition (Exact differential). A differential form $\mathbf{F} \cdot d\mathbf{r}$ is called *exact* if there exists a scalar function f such that $\mathbf{F} = \nabla f$, i.e.,

$$\mathbf{F} \cdot d\mathbf{r} = df = \nabla f \cdot d\mathbf{r} = \frac{\partial f}{\partial x_i} dx_i.$$

The following proposition provides a necessary condition for a differential to be exact.

Proposition (Exactness symmetry condition). If $\mathbf{F} = \nabla f$ for some smooth function f , then

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$$

for all i, j .

Proof. Since $F_i = \frac{\partial f}{\partial x_i}$, the equality of mixed partial derivatives gives

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i} = \frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial F_j}{\partial x_i}. \quad \square$$

Remark. For an exact differential, the fundamental theorem for line integrals can be written as

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C df = f(\mathbf{b}) - f(\mathbf{a}).$$

Differentials satisfy the following algebraic properties, which can be useful for recognizing exact differentials.

Proposition (Properties of differentials). For differentiable functions f and g , and constants $\lambda, \mu \in \mathbb{R}$:

$$\begin{aligned} d(\lambda f + \mu g) &= \lambda df + \mu dg && \text{(linearity)} \\ d(fg) &= g df + f dg && \text{(product rule)} \end{aligned}$$

These rules can sometimes be used to recognize an exact differential and find its potential by inspection.

Example (Exact differential integration). Consider

$$\int_C 3x^2y \sin z \, dx + x^3 \sin z \, dy + x^3y \cos z \, dz.$$

We see that if we integrate the first term with respect to x , we obtain $x^3y \sin z$. We obtain the same thing if we integrate the second and third term. So this is equal to

$$\int_C d(x^3y \sin z) = [x^3y \sin z]_{\mathbf{a}}^{\mathbf{b}}.$$

2.4 Work and potential energy

We now apply the theory of line integrals to mechanics.

Definition (Work). Let $\mathbf{F}(\mathbf{r})$ be a force field. The *work done* by $\mathbf{F}$ on a particle moving along a curve C is

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

This represents the accumulated effect of the force component along the direction of motion.

Definition (Kinetic energy). The *kinetic energy* of a particle of mass m moving with velocity $\dot{\mathbf{r}}$ is

$$T = \frac{1}{2}m|\dot{\mathbf{r}}|^2.$$

Theorem (Work-energy theorem). Consider a particle of mass m moving under a force $\mathbf{F}(\mathbf{r})$ according to Newton's second law, $\mathbf{F} = m\ddot{\mathbf{r}}$. The work done by $\mathbf{F}$ as the particle moves along a curve C from $\mathbf{a} = \mathbf{r}(t_1)$ to $\mathbf{b} = \mathbf{r}(t_2)$ equals the change in kinetic energy:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = T(t_2) - T(t_1).$$

Proof. Differentiating T with respect to time,

$$\frac{dT}{dt} = m\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} = \mathbf{F} \cdot \dot{\mathbf{r}}.$$

Integrating from t_1 to t_2 gives

$$T(t_2) - T(t_1) = \int_{t_1}^{t_2} \mathbf{F} \cdot \dot{\mathbf{r}} \, dt = \int_C \mathbf{F} \cdot d\mathbf{r}. \quad \square$$

Definition (Potential energy). If a force field $\mathbf{F}$ is conservative, so that $\mathbf{F} = -\nabla V$ for some scalar function V , then $V(\mathbf{r})$ is called the *potential energy*. The work done by $\mathbf{F}$ along a curve C from $\mathbf{a}$ to $\mathbf{b}$ is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = - \int_C \nabla V \cdot d\mathbf{r} = V(\mathbf{a}) - V(\mathbf{b}).$$

Theorem (Conservation of energy). For a conservative force $\mathbf{F} = -\nabla V$, the total mechanical energy $E = T + V$ is conserved along the motion of the particle, i.e., E remains constant in time.

Proof. By the work-energy theorem, the change in kinetic energy equals the work done: $\Delta T = \int_C \mathbf{F} \cdot d\mathbf{r}$. For a conservative force, this equals $V(\mathbf{a}) - V(\mathbf{b}) = -\Delta V$. Hence $\Delta T + \Delta V = 0$, so $T + V$ is constant. $\square$

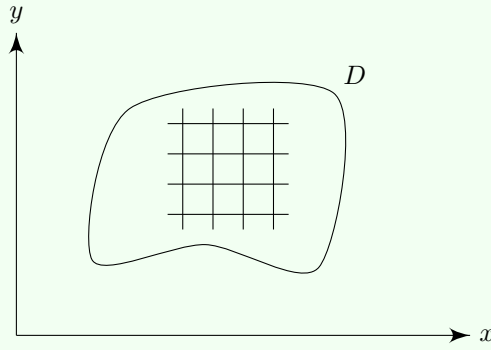
Remark. The converse also holds: if energy is conserved for all motions in a force field, then the force must be conservative.

3 Integration in $\mathbb{R}^2$ and $\mathbb{R}^3$

3.1 Integrals over subsets of $\mathbb{R}^2$

We now extend the notion of integration to functions of two variables, integrating over regions in the plane.

Definition (Area integral). Let $D \subseteq \mathbb{R}^2$ be a bounded region and let $f : D \rightarrow \mathbb{R}$ be a function. To define the integral of f over D , we partition D into N disjoint subsets of simple shapes (e.g., rectangles or triangles), labelled by $I = 1, \dots, N$, with areas δA_I .



To ensure a well-defined limit, we require that each subset can be contained in a disc of diameter ℓ , preventing the subsets from becoming arbitrarily elongated. The *area integral* (or *double integral*) of f over D is defined as

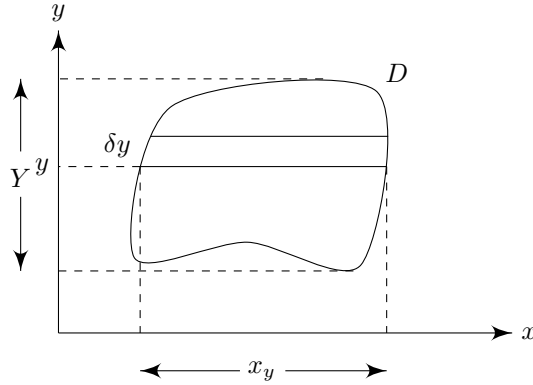
$$\int_D f(\mathbf{r}) \, dA = \lim_{\ell \rightarrow 0} \sum_I f(\mathbf{r}_I) \delta A_I,$$

where $\mathbf{r}_I$ is a point chosen within each subset. The limit is taken as $\ell \rightarrow 0$, $N \rightarrow \infty$, with the union of the subsets approaching D . The integral *exists* if this limit is independent of the choice of partition and sample points.

Remark (Geometric interpretations). If $f = 1$, then $\int_D dA$ equals the area of D . More generally, if $f(x, y) \geq 0$, then $\int_D f \, dA$ represents the volume of the region bounded below by D and above by the surface $z = f(x, y)$.

While the definition permits partitions of arbitrary shape, in practice we use rectangular partitions aligned with the coordinate axes.

To evaluate the integral, we partition D into rectangles of size $\delta A_I = \delta x \delta y$. Consider a horizontal strip of height δy at fixed y . Summing over rectangles in this strip and taking $\delta x \rightarrow 0$ yields a contribution $\delta y \int_{x_y} f(x, y) \, dx$, where $x_y = \{x : (x, y) \in D\}$ denotes the set of x -values for which $(x, y) \in D$.



We sum over all such strips and take $\delta y \rightarrow 0$, giving

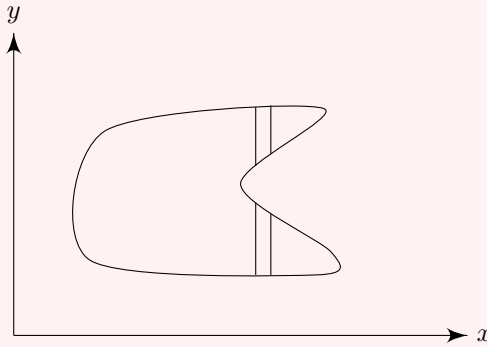
Proposition (Iterated integral formula).

$$\int_D f(x, y) \, dA = \int_Y \left(\int_{x_y} f(x, y) \, dx \right) dy,$$

where Y is the range of y -values in D and $x_y = \{x : (x, y) \in D\}$.

Remark (Disconnected integration domains). The set x_y may consist of multiple disjoint intervals if D is non-convex. For example, if $x_y = [a_1, b_1] \cup [a_2, b_2]$, then

$$\int_{x_y} f(x) \, dx = \int_{a_1}^{b_1} f(x) \, dx + \int_{a_2}^{b_2} f(x) \, dx.$$



Alternatively, we may integrate in the opposite order, first over y and then over x :

$$\int_D f(x, y) \, dA = \int_X \left(\int_{y_x} f(x, y) \, dy \right) dx,$$

where $y_x = \{y : (x, y) \in D\}$. The following theorem guarantees that both orders yield the same result.

Theorem (Fubini's theorem). Let $f : D \rightarrow \mathbb{R}$ be a continuous function and let $D \subseteq \mathbb{R}^2$ be a compact (closed and bounded) region. Then the order of integration may be interchanged:

$$\iint_D f \, dx \, dy = \iint_D f \, dy \, dx.$$

Remark. The hypotheses of Fubini's theorem can be relaxed significantly; the theorem holds under much weaker conditions, though verification requires care in specific cases.

Remark (Area element). In Cartesian coordinates, the *area element* is $dA = dx \, dy$.

Example (Triangle area integral). We integrate over the triangle bounded by $(0,0)$, $(2,0)$ and $(0,1)$. We want to integrate the function $f(x,y) = x^2y$ over the area. So

$$\begin{aligned} \int_D f(x,y) \, dA &= \int_0^1 \left(\int_0^{2-2y} x^2y \, dx \right) dy \\ &= \int_0^1 y \left[\frac{x^3}{3} \right]_0^{2-2y} dy \\ &= \frac{8}{3} \int_0^1 y(1-y)^3 dy \\ &= \frac{2}{15} \end{aligned}$$

We can integrate it the other way round:

$$\begin{aligned} \int_D x^2y \, dA &= \int_0^2 \int_0^{1-x/2} x^2y \, dy \, dx \\ &= \int_0^2 x^2 \left[\frac{1}{2}y^2 \right]_0^{1-x/2} dx \\ &= \int_0^2 \frac{x^2}{2} \left(1 - \frac{x}{2} \right)^2 dx \\ &= \frac{2}{15} \end{aligned}$$

Remark. By Fubini's theorem, if one order of integration proves difficult, we may attempt the other order, which may yield a simpler computation.

A special case where double integrals simplify considerably is when the integrand factors.

Definition (Separable function). A function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is called *separable* if it can be written as a product $f(x, y) = g(x)h(y)$ for some functions g and h of one variable each.

Proposition (Separable integral factorization). Let $f(x, y) = g(x)h(y)$ be a separable function and let $D = [a, b] \times [c, d]$ be a rectangle. Then

$$\int_D f(x, y) \, dx \, dy = \left(\int_a^b g(x) \, dx \right) \left(\int_c^d h(y) \, dy \right).$$

3.2 Change of variables for an integral in $\mathbb{R}^2$

Just as for single-variable integrals, a suitable change of variables can greatly simplify the evaluation of a double integral.

Theorem (Change of variables for area integrals). Let $(x, y) = \Phi(u, v)$ be a smooth, invertible transformation mapping a region D' in the (u, v) -plane to a region D in the (x, y) -plane. Then

$$\int_D f(x, y) \, dx \, dy = \int_{D'} f(x(u, v), y(u, v)) |J| \, du \, dv,$$

where the *Jacobian* J of the transformation is

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

Equivalently, the area elements are related by $dx \, dy = |J| \, du \, dv$.

Proof. Note that Φ maps from (u, v) to (x, y) . Consider a small rectangle in the (u, v) -plane with sides δu and δv . By Taylor's theorem,

$$\delta x = x(u + \delta u, v + \delta v) - x(u, v) \approx \frac{\partial x}{\partial u} \delta u + \frac{\partial x}{\partial v} \delta v,$$

and similarly for δy . In matrix form,

$$\begin{pmatrix} \delta x \\ \delta y \end{pmatrix} \approx \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} \begin{pmatrix} \delta u \\ \delta v \end{pmatrix}.$$

Since a linear map scales areas by the absolute value of its determinant, the area element $\delta u \, \delta v$ is mapped to $|J| \, \delta u \, \delta v$. Hence

$$dx \, dy = |J| \, du \, dv. \quad \square$$

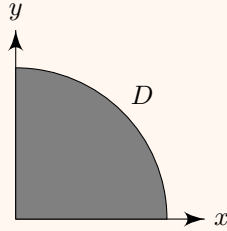
Example (Polar area element derivation). We transform from (x, y) to (ρ, φ) with

$$\begin{aligned}x &= \rho \cos \varphi \\y &= \rho \sin \varphi\end{aligned}$$

We have previously calculated that $|J| = \rho$. So

$$dA = \rho \, d\rho \, d\varphi.$$

Suppose we want to integrate a function over the quarter disc D of radius R in the first quadrant.



Let the function to be integrated be $f = \exp(-(x^2 + y^2)/2) = \exp(-\rho^2/2)$. Then

$$\begin{aligned}\int f \, dA &= \int f \rho \, d\rho \, d\varphi \\&= \int_{\rho=0}^R \left(\int_{\varphi=0}^{\pi/2} e^{-\rho^2/2} \rho \, d\varphi \right) d\rho\end{aligned}$$

Note that in polar coordinates, we are integrating over a rectangle and the function is separable. So this is equal to

$$\begin{aligned}&= \left[-e^{-\rho^2/2} \right]_0^R [\varphi]_0^{\pi/2} \\&= \frac{\pi}{2} \left(1 - e^{-R^2/2} \right).\end{aligned}\tag{*}$$

Note that the integral exists as $R \rightarrow \infty$, giving $\frac{\pi}{2}$.

Now we evaluate the same integral over the first quadrant in Cartesian coordinates. Since f is separable,

$$\int_{x=0}^{\infty} \int_{y=0}^{\infty} e^{-(x^2+y^2)/2} \, dx \, dy = \left(\int_0^{\infty} e^{-x^2/2} \, dx \right) \left(\int_0^{\infty} e^{-y^2/2} \, dy \right) = \frac{\pi}{2},$$

where the last equality follows from (*). So each factor must equal $\sqrt{\pi/2}$, i.e.

$$\int_0^{\infty} e^{-x^2/2} \, dx = \sqrt{\frac{\pi}{2}}.$$

3.3 Generalization to $\mathbb{R}^3$

The definition of the area integral extends naturally to three dimensions.

Definition (Volume integral). Let $V \subseteq \mathbb{R}^3$ be a bounded region and let $f : V \rightarrow \mathbb{R}$ be a function. We partition V into N disjoint subsets (e.g., cuboids) labelled by I , each with volume δV_I and contained within a sphere of diameter ℓ . The *volume integral* (or *triple integral*) of f over V is

$$\int_V f(\mathbf{r}) \, dV = \lim_{\ell \rightarrow 0} \sum_I f(\mathbf{r}_I^*) \delta V_I,$$

where $\mathbf{r}_I^*$ is a point chosen within each subset and the limit is taken as $\ell \rightarrow 0$, $N \rightarrow \infty$, with the union of subsets approaching V .

To evaluate the integral, we use iterated integration. Taking $\delta V_I = \delta x \delta y \delta z$ and integrating successively, we obtain

$$\int_V f(\mathbf{r}) \, dV = \int_D \left(\int_{Z_{xy}} f(x, y, z) \, dz \right) dx \, dy,$$

where Z_{xy} denotes the range of z -values for each (x, y) in the projection D of V onto the (x, y) -plane.

Alternatively, we may integrate over a cross-sectional area first:

$$\int_V f(\mathbf{r}) \, dV = \int_Z \left(\int_{D_z} f(x, y, z) \, dx \, dy \right) dz,$$

where D_z is the cross-section of V at height z .

Remark (Physical interpretation). If $f = 1$, the volume integral gives the volume of V . More generally, if $\rho(\mathbf{r})$ represents a density (e.g., mass density, charge density, or probability density), then $\int_V \rho(\mathbf{r}) \, dV$ gives the total quantity contained in V .

Definition (Volume element). The *volume element* is dV .

Proposition (Cartesian volume element). $dV = dx \, dy \, dz$.

As in two dimensions, a change of variables transforms the volume element by the absolute value of the Jacobian.

Theorem (Change of variables for volume integrals). Let $(x, y, z) = \Phi(u, v, w)$ be a smooth, invertible transformation mapping a region V' in (u, v, w) -space to a region V in (x, y, z) -space. Then

$$\int_V f \, dx \, dy \, dz = \int_{V'} f |J| \, du \, dv \, dw,$$

where the Jacobian is

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

Proposition (Cylindrical and spherical volume elements). In cylindrical coordinates (ρ, φ, z) , the volume element is

$$dV = \rho d\rho d\varphi dz.$$

In spherical coordinates (r, θ, φ) , the volume element is

$$dV = r^2 \sin \theta dr d\theta d\varphi.$$

Proof. These follow from computing the Jacobians of the coordinate transformations given in Section 1. $\square$

Example (Spherical symmetry volume integral). Suppose $f(\mathbf{r})$ is spherically symmetric and V is a sphere of radius a centered on the origin. Then

$$\begin{aligned} \int_V f dV &= \int_{r=0}^a \int_{\theta=0}^{\pi} \int_{\varphi=0}^{2\pi} f(r) r^2 \sin \theta dr d\theta d\varphi \\ &= \int_0^a dr \int_0^{\pi} d\theta \int_0^{2\pi} d\varphi r^2 f(r) \sin \theta \\ &= \int_0^a r^2 f(r) dr \left[-\cos \theta \right]_0^{\pi} \left[\varphi \right]_0^{2\pi} \\ &= 4\pi \int_0^a f(r) r^2 dr, \end{aligned}$$

where in the second line we placed the differentials next to the integral signs, with the limits now implicit in the order of integration. The integral separates since f depends only on r .

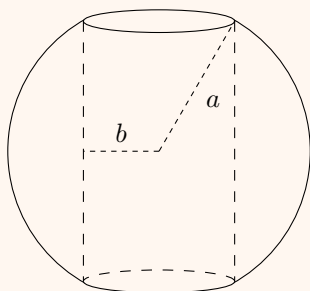
This is a useful general result: the integral of a spherically symmetric function over a ball reduces to a one-dimensional integral, which can be interpreted as summing over spherical shells of thickness δr and volume $4\pi r^2 \delta r$.

Setting $f = 1$ gives the volume of a sphere as $\frac{4}{3}\pi a^3$.

Example (Sphere with cylinder removed). Consider the region within a

sphere of radius a with a cylinder of radius b ($b < a$) removed, defined by

$$\begin{aligned}x^2 + y^2 + z^2 &\leq a^2 \\x^2 + y^2 &\geq b^2.\end{aligned}$$



We use cylindrical coordinates. The second criterion gives

$$b \leq \rho \leq a.$$

For the $x^2 + y^2 + z^2 \leq a^2$ criterion, we have

$$-\sqrt{a^2 - \rho^2} \leq z \leq \sqrt{a^2 - \rho^2}.$$

So the volume is

$$\begin{aligned}\int_V dV &= \int_b^a d\rho \int_0^{2\pi} d\varphi \int_{-\sqrt{a^2 - \rho^2}}^{\sqrt{a^2 - \rho^2}} dz \rho \\&= 2\pi \int_b^a 2\rho \sqrt{a^2 - \rho^2} d\rho \\&= 2\pi \left[-\frac{2}{3}(a^2 - \rho^2)^{3/2} \right]_b^a \\&= \frac{4}{3}\pi(a^2 - b^2)^{3/2}.\end{aligned}$$

Example (Hemisphere total charge). Suppose the density of electric charge is $\rho(\mathbf{r}) = \rho_0 \frac{z}{a}$ in a hemisphere H of radius a , with $z \geq 0$. What is the total charge of H ?

In spherical polar coordinates, the region is described by

$$r \leq a, \quad 0 \leq \varphi \leq 2\pi, \quad 0 \leq \theta \leq \frac{\pi}{2},$$

and the charge density becomes

$$\rho(\mathbf{r}) = \frac{\rho_0}{a} r \cos \theta.$$

The total charge Q in H is

$$\begin{aligned}\int_H \rho \, dV &= \int_0^a dr \int_0^{\pi/2} d\theta \int_0^{2\pi} d\varphi \frac{\rho_0}{a} r \cos \theta r^2 \sin \theta \\ &= \frac{\rho_0}{a} \int_0^a r^3 \, dr \int_0^{\pi/2} \sin \theta \cos \theta \, d\theta \int_0^{2\pi} d\varphi \\ &= \frac{\rho_0}{a} \left[\frac{r^4}{4} \right]_0^a \left[\frac{1}{2} \sin^2 \theta \right]_0^{\pi/2} [\varphi]_0^{2\pi} \\ &= \frac{\rho_0 \pi a^3}{4}.\end{aligned}$$

3.4 Further generalizations

Integration in $\mathbb{R}^n$

The definitions and results above extend naturally to n dimensions. For a region $D \subseteq \mathbb{R}^n$ and a function $f : D \rightarrow \mathbb{R}$, the n -dimensional integral is

$$\int_D f(x_1, \dots, x_n) \, dx_1 \cdots dx_n.$$

Theorem (Change of variables in $\mathbb{R}^n$). Let $\mathbf{x} = \Phi(\mathbf{u})$ be a smooth, invertible transformation from $D' \subseteq \mathbb{R}^n$ to $D \subseteq \mathbb{R}^n$. Then

$$\int_D f(x_1, \dots, x_n) \, dx_1 \cdots dx_n = \int_{D'} f(x_1(\mathbf{u}), \dots, x_n(\mathbf{u})) |J| \, du_1 \cdots du_n,$$

where $J = \det \left(\frac{\partial x_i}{\partial u_j} \right)$ is the Jacobian of the transformation.

Change of variables for $n = 1$

In single-variable calculus, the change of variables formula takes a slightly different form due to the convention for definite integrals. For a transformation $x = x(u)$ mapping an interval to an interval, we have

$$\int_a^b f(x) \, dx = \int_\alpha^\beta f(x(u)) \frac{dx}{du} \, du,$$

where $x(\alpha) = a$ and $x(\beta) = b$. Unlike in higher dimensions, we do not take the absolute value of the Jacobian; instead, the sign of $\frac{dx}{du}$ accounts for the orientation of the integral.

Remark. If we wish to write the integral with the limits in standard order (smaller limit below), we can use

$$\int_D f(x) \, dx = \int_{D'} f(x(u)) \left| \frac{dx}{du} \right| \, du,$$

where the absolute value compensates for a possible reversal of orientation.

This form is consistent with the higher-dimensional formula but is not the standard convention in single-variable calculus.

Vector-valued integrals

The integral of a vector-valued function $\mathbf{F} : V \rightarrow \mathbb{R}^n$ over a region V is defined componentwise.

Definition (Vector-valued integral). Let $\mathbf{F}(\mathbf{r}) = F_i(\mathbf{r})\mathbf{e}_i$ be a vector field. The integral of $\mathbf{F}$ over a region V is

$$\int_V \mathbf{F}(\mathbf{r}) \, dV = \left(\int_V F_i(\mathbf{r}) \, dV \right) \mathbf{e}_i.$$

Example (Center of mass). For a body occupying a region V with mass density $\rho(\mathbf{r})$, the total mass is

$$M = \int_V \rho(\mathbf{r}) \, dV,$$

and the center of mass is

$$\mathbf{R} = \frac{1}{M} \int_V \mathbf{r} \rho(\mathbf{r}) \, dV.$$

Example (Hemisphere center of mass). Consider a solid hemisphere H with $r \leq a$, $z \geq 0$ with uniform density ρ . The mass is

$$M = \int_H \rho \, dV = \frac{2}{3} \pi a^3 \rho.$$

Writing $\mathbf{R} = (X, Y, Z)$, by symmetry we expect $X = Y = 0$. To verify,

$$\begin{aligned} X &= \frac{1}{M} \int_H x \rho \, dV \\ &= \frac{\rho}{M} \int_0^a \int_0^{\pi/2} \int_0^{2\pi} x r^2 \sin \theta \, d\varphi \, d\theta \, dr \\ &= \frac{\rho}{M} \int_0^a r^3 \, dr \int_0^{\pi/2} \sin^2 \theta \, d\theta \int_0^{2\pi} \cos \varphi \, d\varphi \\ &= 0, \end{aligned}$$

since $\int_0^{2\pi} \cos \varphi \, d\varphi = 0$. Similarly, $Y = 0$.

For Z , we compute

$$\begin{aligned} Z &= \frac{\rho}{M} \int_0^a r^3 \, dr \int_0^{\pi/2} \sin \theta \cos \theta \, d\theta \int_0^{2\pi} d\varphi \\ &= \frac{\rho}{M} \left[\frac{a^4}{4} \right] \left[\frac{1}{2} \sin^2 \theta \right]_0^{\pi/2} 2\pi \\ &= \frac{3a}{8}. \end{aligned}$$

So $\mathbf{R} = (0, 0, 3a/8)$.

4 Surfaces and surface integrals

4.1 Surfaces and normals

We now turn to the study of surfaces in $\mathbb{R}^3$. A surface can be specified implicitly by an equation of the form $f(\mathbf{r}) = c$, where $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a smooth function and c is a constant. For example, the equation $x^2 + y^2 + z^2 = 1$ defines the unit sphere.

To find the normal to such a surface, consider any curve $\mathbf{r}(u)$ lying entirely on the surface S . Since $f(\mathbf{r}(u)) = c$ along the curve, differentiating with respect to u gives

$$\frac{d}{du}[f(\mathbf{r}(u))] = \nabla f \cdot \frac{d\mathbf{r}}{du} = 0.$$

This shows that ∇f is perpendicular to the tangent vector $\frac{d\mathbf{r}}{du}$. Since this holds for every curve on S , we conclude:

Proposition (Gradient as surface normal). Let S be a surface defined by $f(\mathbf{r}) = c$, where f is smooth and $\nabla f \neq \mathbf{0}$ on S . Then ∇f is normal to S at each point.

Example (Level surfaces: sphere and hyperboloid).

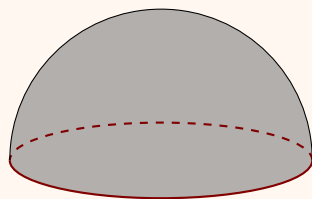
- (i) Take the sphere $f(\mathbf{r}) = x^2 + y^2 + z^2 = c$ for $c > 0$. Then $\nabla f = 2(x, y, z) = 2\mathbf{r}$, which is clearly normal to the sphere.
- (ii) Take $f(\mathbf{r}) = x^2 + y^2 - z^2 = c$, which is a hyperboloid. Then $\nabla f = 2(x, y, -z)$.

In the special case $c = 0$, we have a double cone with a singularity at the apex $\mathbf{0}$. Here $\nabla f = \mathbf{0}$, and we cannot define a meaningful normal direction.

Definition (Boundary of a surface). The *boundary* of a surface S , denoted ∂S , is the curve (or union of curves) forming the edge of S . For example, if the upper hemisphere is defined by $x^2 + y^2 + z^2 = a^2$ with $z \geq 0$, then ∂S is the circle $x^2 + y^2 = a^2$ in the plane $z = 0$.

Definition (Bounded and closed surfaces). A surface is *bounded* if it can be contained within a sphere of finite radius, and *unbounded* otherwise. A bounded surface with no boundary (i.e., $\partial S = \emptyset$) is called *closed*. For example, a sphere is closed, while a hemisphere is bounded but not closed.

Example (Hemisphere boundary).



The boundary of a hemisphere is a circle (drawn in red).

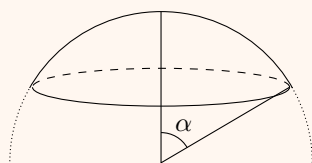
Definition (Orientable surface). At each point of a smooth surface S , there are exactly two unit normal vectors, differing only in sign. A surface is *orientable* if it is possible to choose a unit normal $\mathbf{n}$ at each point that varies continuously over the entire surface. Such a choice is called an *orientation* of the surface.

Remark. Most surfaces encountered in practice are orientable. For a sphere, one can choose the outward-pointing normal everywhere; for a bounded surface with boundary, one typically speaks of “upper” and “lower” or “inward” and “outward” orientations. The Möbius strip and Klein bottle are examples of non-orientable surfaces.

4.2 Parametrized surfaces and area

While the implicit representation $f(\mathbf{r}) = c$ is useful for some purposes, it is often more convenient to describe a surface parametrically. A *parametrization* of a surface S is a smooth function $\mathbf{r} : D \rightarrow \mathbb{R}^3$, where $D \subseteq \mathbb{R}^2$ is a region in the (u, v) -plane, such that the image of $\mathbf{r}$ is S . Each point on S is then labelled by coordinates (u, v) .

Example (Spherical cap parametrization). Let S be part of a sphere of radius a with $0 \leq \theta \leq \alpha$.



We can then label the points on the sphere by the angles θ, φ , with

$$\mathbf{r}(\theta, \varphi) = (a \sin \theta \cos \varphi, a \sin \theta \sin \varphi, a \cos \theta) = a \mathbf{e}_r.$$

We restrict the values of θ, φ by $0 \leq \theta \leq \alpha$, $0 \leq \varphi \leq 2\pi$, so that each point is only covered once.

Remark. A complete specification of a parametrized surface requires both the function $\mathbf{r}(u, v)$ and the domain $D \subseteq \mathbb{R}^2$ of allowed parameter values. The domain D determines the bounds of integration when computing surface integrals.

For the parametrization to describe a genuine two-dimensional surface, we require that $\mathbf{r}$ depend non-degenerately on both parameters. The partial derivatives $\frac{\partial \mathbf{r}}{\partial u}$ and $\frac{\partial \mathbf{r}}{\partial v}$ are tangent vectors to the coordinate curves on S (curves with v constant and u constant, respectively). For the surface to be non-degenerate, these tangent vectors must be linearly independent.

Definition (Regular parametrization). A parametrization $\mathbf{r} : D \rightarrow \mathbb{R}^3$ is called *regular* if

$$\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \neq \mathbf{0}$$

at every point $(u, v) \in D$, i.e., the tangent vectors are never parallel. All parametrizations in this course are assumed to be regular.

We now derive the formula for the area of a parametrized surface. Consider a small rectangle in the parameter domain D with sides δu and δv . Under the parametrization, this maps to a small parallelogram on S with sides approximately $\frac{\partial \mathbf{r}}{\partial u} \delta u$ and $\frac{\partial \mathbf{r}}{\partial v} \delta v$. The area of this parallelogram is the magnitude of the cross product:

$$\delta S = \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| \delta u \delta v.$$

The cross product itself gives the *vector area element*, which includes information about the orientation:

$$\delta \mathbf{S} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \delta u \delta v = \mathbf{n} \delta S,$$

where $\mathbf{n}$ is the unit normal. The ordering of u and v determines the sign (orientation) of the normal.

Taking the limit as $\delta u, \delta v \rightarrow 0$, we obtain:

Proposition (Parametrized surface area elements). The *vector area element* is

$$d\mathbf{S} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} du dv.$$

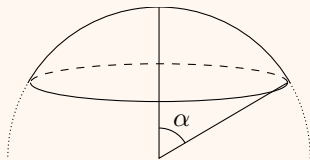
The *scalar area element* is

$$dS = \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv.$$

By summing and taking limits, the area of S is

$$\int_S dS = \int_D \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv.$$

Example (Spherical cap scalar area). Consider again the part of the sphere of radius a with $0 \leq \theta \leq \alpha$.



The parametrization is $\mathbf{r}(\theta, \varphi) = (a \sin \theta \cos \varphi, a \sin \theta \sin \varphi, a \cos \theta) = a \mathbf{e}_r$, so

$$\frac{\partial \mathbf{r}}{\partial \theta} = a \mathbf{e}_\theta, \quad \frac{\partial \mathbf{r}}{\partial \varphi} = a \sin \theta \mathbf{e}_\varphi,$$

and hence

$$\frac{\partial \mathbf{r}}{\partial \theta} \times \frac{\partial \mathbf{r}}{\partial \varphi} = a^2 \sin \theta \mathbf{e}_r, \quad dS = a^2 \sin \theta \, d\theta \, d\varphi.$$

With bounds $0 \leq \theta \leq \alpha$ and $0 \leq \varphi \leq 2\pi$, the area is

$$\int_0^{2\pi} \int_0^\alpha a^2 \sin \theta \, d\theta \, d\varphi = 2\pi a^2 (1 - \cos \alpha).$$

4.3 Surface integral of vector fields

Having defined the area element on a surface, we now consider integrals of vector fields over surfaces. Given a surface S parametrized by $\mathbf{r}(u, v)$ and a vector field $\mathbf{F}(\mathbf{r})$, we wish to measure the total “flow” of $\mathbf{F}$ through S .

A naive approach might be to integrate $|\mathbf{F}|$ over S , but this does not capture the physics correctly: if $\mathbf{F}$ is tangent to the surface everywhere, nothing actually passes through, yet $|\mathbf{F}|$ may be large. The correct quantity to integrate is the component of $\mathbf{F}$ normal to the surface.

Definition (Surface integral of a vector field). Let S be an oriented surface parametrized by $\mathbf{r} : D \rightarrow \mathbb{R}^3$ and let $\mathbf{F}$ be a vector field. The *surface integral* (or *flux*) of $\mathbf{F}$ through S is

$$\int_S \mathbf{F} \cdot d\mathbf{S} = \int_S \mathbf{F} \cdot \mathbf{n} \, dS = \int_D \mathbf{F}(\mathbf{r}(u, v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du \, dv.$$

This measures the net rate at which $\mathbf{F}$ crosses the surface in the direction of the chosen normal.

Remark (Parametrization independence). For a given orientation, the flux integral is independent of the choice of parametrization. Reversing the orientation (i.e., reversing the direction of $\mathbf{n}$) changes the sign of the integral. This is equivalent to interchanging the order of u and v in the parametrization.

Example (Sphere vector area element). Consider a sphere of radius a , $\mathbf{r}(\theta, \varphi)$. Then

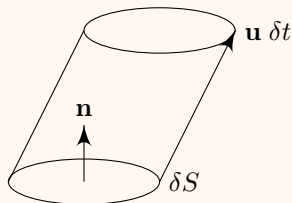
$$\frac{\partial \mathbf{r}}{\partial \theta} = a \mathbf{e}_\theta, \quad \frac{\partial \mathbf{r}}{\partial \varphi} = a \sin \theta \mathbf{e}_\varphi.$$

The vector area element is

$$d\mathbf{S} = a^2 \sin \theta \mathbf{e}_r \, d\theta \, d\varphi,$$

taking the outward normal $\mathbf{n} = \mathbf{e}_r = \mathbf{r}/a$.

Suppose we want to calculate the fluid flux through the surface. The *velocity field* $\mathbf{u}(\mathbf{r})$ of a fluid gives the motion of a small volume of fluid $\mathbf{r}$. Assume that $\mathbf{u}$ depends smoothly on $\mathbf{r}$ (and t). For any small area δS , on a surface S , the volume of fluid crossing it in time δt is $\mathbf{u} \cdot d\mathbf{S} \, \delta t$.



So the amount of flow of $\mathbf{u}$ over at time δt through S is

$$\delta t \int_S \mathbf{u} \cdot d\mathbf{S}.$$

So $\int_S \mathbf{u} \cdot d\mathbf{S}$ is the *rate* of volume crossing S .

For example, let $\mathbf{u} = (-x, 0, z)$ and S be the section of a sphere of radius a with $0 \leq \varphi \leq 2\pi$ and $0 \leq \theta \leq \alpha$. Then

$$d\mathbf{S} = a^2 \sin \theta \mathbf{n} \, d\varphi \, d\theta,$$

with

$$\mathbf{n} = \frac{\mathbf{r}}{a} = \frac{1}{a}(x, y, z).$$

So

$$\mathbf{n} \cdot \mathbf{u} = \frac{1}{a}(-x^2 + z^2) = a(-\sin^2 \theta \cos^2 \varphi + \cos^2 \theta).$$

Therefore

$$\begin{aligned} \int_S \mathbf{u} \cdot d\mathbf{S} &= \int_0^\alpha \int_0^{2\pi} a^3 \sin \theta [(\cos^2 \theta - 1) \cos^2 \varphi + \cos^2 \theta] \, d\varphi \, d\theta \\ &= \int_0^\alpha a^3 \sin \theta [\pi(\cos^2 \theta - 1) + 2\pi \cos^2 \theta] \, d\theta \\ &= \int_0^\alpha a^3 \pi (3 \cos^2 \theta - 1) \sin \theta \, d\theta \\ &= \pi a^3 [\cos \theta - \cos^3 \theta]_0^\alpha \\ &= \pi a^3 \cos \alpha \sin^2 \alpha. \end{aligned}$$

We now verify that the area element is independent of the choice of parametrization. Let $\mathbf{r}(u, v)$ and $\mathbf{r}(\tilde{u}, \tilde{v})$ be two regular parametrizations for the same surface. By the chain rule,

$$\begin{aligned}\frac{\partial \mathbf{r}}{\partial u} &= \frac{\partial \mathbf{r}}{\partial \tilde{u}} \frac{\partial \tilde{u}}{\partial u} + \frac{\partial \mathbf{r}}{\partial \tilde{v}} \frac{\partial \tilde{v}}{\partial u}, \\ \frac{\partial \mathbf{r}}{\partial v} &= \frac{\partial \mathbf{r}}{\partial \tilde{u}} \frac{\partial \tilde{u}}{\partial v} + \frac{\partial \mathbf{r}}{\partial \tilde{v}} \frac{\partial \tilde{v}}{\partial v}.\end{aligned}$$

Taking the cross product yields

$$\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} = \frac{\partial(\tilde{u}, \tilde{v})}{\partial(u, v)} \frac{\partial \mathbf{r}}{\partial \tilde{u}} \times \frac{\partial \mathbf{r}}{\partial \tilde{v}},$$

where $\frac{\partial(\tilde{u}, \tilde{v})}{\partial(u, v)}$ is the Jacobian of the coordinate transformation. Since the area elements in parameter space are related by

$$d\tilde{u} d\tilde{v} = \frac{\partial(\tilde{u}, \tilde{v})}{\partial(u, v)} du dv,$$

we obtain the following result.

Proposition (Parametrization independence of area elements). The scalar area element is independent of the choice of parametrization:

$$dS = \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv = \left| \frac{\partial \mathbf{r}}{\partial \tilde{u}} \times \frac{\partial \mathbf{r}}{\partial \tilde{v}} \right| d\tilde{u} d\tilde{v}.$$

The vector area element

$$d\mathbf{S} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} du dv = \frac{\partial \mathbf{r}}{\partial \tilde{u}} \times \frac{\partial \mathbf{r}}{\partial \tilde{v}} d\tilde{u} d\tilde{v}$$

is independent of parametrization provided (u, v) and $(\tilde{u}, \tilde{v})$ induce the same orientation on the surface.

4.4 Change of variables in $\mathbb{R}^2$ and $\mathbb{R}^3$ revisited

The change of variables formula for integrals can be derived using the geometric ideas developed above. We present an alternative derivation that unifies the two- and three-dimensional cases.

Change of variable formula in $\mathbb{R}^2$

The two-dimensional change of variables formula can be derived from the surface area formula by embedding $\mathbb{R}^2$ in $\mathbb{R}^3$.

Consider a region $S \subset \mathbb{R}^2$ with a parametrization $(x(u, v), y(u, v))$ for $(u, v) \in D$. We embed this in $\mathbb{R}^3$ by setting $\mathbf{r}(u, v) = (x(u, v), y(u, v), 0)$. Computing the cross product gives

$$\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} = (0, 0, J),$$

where $J = \frac{\partial(x,y)}{\partial(u,v)}$ is the Jacobian of the coordinate transformation. Thus $|\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}| = |J|$, and from the surface integral formula we obtain

$$\int_S f(x, y) \, dS = \int_D f(x(u, v), y(u, v)) |J| \, du \, dv,$$

which is the standard change of variables formula in $\mathbb{R}^2$.

Change of variable formula in $\mathbb{R}^3$

For a volume integral in $\mathbb{R}^3$, we proceed similarly using the scalar triple product. Let V be a region parametrized by $\mathbf{r}(u, v, w)$ for $(u, v, w) \in D$. A small displacement in parameter space gives

$$\delta \mathbf{r} = \frac{\partial \mathbf{r}}{\partial u} \delta u + \frac{\partial \mathbf{r}}{\partial v} \delta v + \frac{\partial \mathbf{r}}{\partial w} \delta w + o(\delta u, \delta v, \delta w).$$

A small cuboid with sides δu , δv , δw in parameter space maps to a parallelepiped with edges $\frac{\partial \mathbf{r}}{\partial u} \delta u$, $\frac{\partial \mathbf{r}}{\partial v} \delta v$, and $\frac{\partial \mathbf{r}}{\partial w} \delta w$. The volume of this parallelepiped is the absolute value of the scalar triple product:

$$\delta V = \left| \frac{\partial \mathbf{r}}{\partial u} \cdot \left(\frac{\partial \mathbf{r}}{\partial v} \times \frac{\partial \mathbf{r}}{\partial w} \right) \right| \delta u \, \delta v \, \delta w = |J| \, \delta u \, \delta v \, \delta w,$$

where $J = \frac{\partial(x,y,z)}{\partial(u,v,w)}$ is the Jacobian. Taking the limit yields $dV = |J| \, du \, dv \, dw$.

5 Geometry of curves and surfaces

We now study the intrinsic geometry of curves and surfaces, characterizing how they bend and twist in space.

5.1 Curvature of curves

Let $\mathbf{r}(s)$ be a curve parametrized by arclength s . The unit tangent vector is $\mathbf{t}(s) = \frac{d\mathbf{r}}{ds}$. Since $\mathbf{t} \cdot \mathbf{t} = 1$, differentiating with respect to s gives $\mathbf{t} \cdot \mathbf{t}' = 0$. Thus $\mathbf{t}'$ is perpendicular to $\mathbf{t}$, i.e., $\mathbf{t}'$ is normal to the curve whenever $\mathbf{t}' \neq \mathbf{0}$.

Definition (Principal normal and curvature). Write $\mathbf{t}' = \kappa \mathbf{n}$, where $\mathbf{n}$ is a unit vector and $\kappa > 0$. Then $\mathbf{n}(s)$ is called the *principal normal* and $\kappa(s)$ is called the *curvature*.

Remark. The curvature κ is defined with respect to the arclength parametrization. If the curve is given in another parametrization, one must either reparametrize by arclength or use the chain rule to express derivatives with respect to s .

To understand the geometric meaning of curvature, we expand the curve in a Taylor series about a point. Taking $s = 0$ at the point of interest, we have

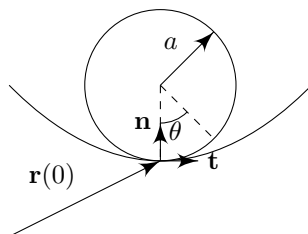
$$\mathbf{r}(s) = \mathbf{r}(0) + s\mathbf{r}'(0) + \frac{1}{2}s^2\mathbf{r}''(0) + O(s^3).$$

Since $\mathbf{r}' = \mathbf{t}$ and $\mathbf{r}'' = \mathbf{t}' = \kappa \mathbf{n}$, this becomes

$$\mathbf{r}(s) = \mathbf{r}(0) + s\mathbf{t}(0) + \frac{1}{2}\kappa(0)s^2\mathbf{n}(0) + O(s^3).$$

This expansion shows that the curve deviates from a straight line (the tangent) quadratically in s , with the coefficient κ controlling the rate of deviation in the direction $\mathbf{n}$.

To interpret κ geometrically, we compare the curve to a circle. Intuitively, a more “curved” curve should be approximated by a circle of smaller radius. Consider a circle of radius a passing through $\mathbf{r}(0)$, lying in the plane spanned by $\mathbf{t}$ and $\mathbf{n}$:



The parametric equation for this circle is

$$\mathbf{r} = \mathbf{r}(0) + a(1 - \cos \theta)\mathbf{n} + a \sin \theta \mathbf{t}.$$

Expanding in powers of θ gives

$$\mathbf{r} = \mathbf{r}(0) + a\theta\mathbf{t} + \frac{1}{2}a\theta^2\mathbf{n} + O(\theta^3).$$

Since the arclength along the circle is $s = a\theta$, we can write $\theta = s/a$, yielding

$$\mathbf{r} = \mathbf{r}(0) + s\mathbf{t} + \frac{1}{2a}s^2\mathbf{n} + O(s^3).$$

Comparing with the Taylor expansion of the curve, we see that the curve and circle agree to second order when $\kappa = 1/a$.

Definition (Osculating circle and radius of curvature). The *osculating circle* at a point on a curve is the circle that best approximates the curve at that point (agreeing to second order in arclength). The *radius of curvature* is the radius of the osculating circle, given by $1/\kappa$.

Proposition (Curvature and radius of curvature). The curvature κ at a point on a curve equals the reciprocal of the radius of the osculating circle at that point: $\kappa = 1/a$.

5.2 The Frenet–Serret frame

In three dimensions, the vectors $\mathbf{t}(s)$ and $\mathbf{n}(s)$ can be completed to an orthonormal basis by adding a third vector.

Definition (Binormal). The *binormal* of a curve is the unit vector $\mathbf{b} = \mathbf{t} \times \mathbf{n}$.

The ordered triple $(\mathbf{t}, \mathbf{n}, \mathbf{b})$ forms a right-handed orthonormal basis at each point of the curve, called the *Frenet–Serret frame*. This frame moves along the curve with the parameter s .

Just as curvature measures how $\mathbf{t}$ changes along the curve, we can measure how $\mathbf{b}$ changes.

Definition (Torsion). The *torsion* $\tau(s)$ is defined by $\mathbf{b}' = -\tau\mathbf{n}$.

Remark. The definition is consistent because $\mathbf{b}'$ must be parallel to $\mathbf{n}$. To see this, note that $\mathbf{b}' = \mathbf{t}' \times \mathbf{n} + \mathbf{t} \times \mathbf{n}' = \mathbf{t} \times \mathbf{n}'$ (since $\mathbf{t}' = \kappa\mathbf{n}$ is parallel to $\mathbf{n}$), so $\mathbf{b}'$ is perpendicular to $\mathbf{t}$. Also, $\mathbf{b} \cdot \mathbf{b} = 1$ implies $\mathbf{b} \cdot \mathbf{b}' = 0$, so $\mathbf{b}'$ is perpendicular to $\mathbf{b}$. Thus $\mathbf{b}'$ must be parallel to $\mathbf{n}$.

The geometry of a space curve is completely determined by how the Frenet–Serret frame $(\mathbf{t}, \mathbf{n}, \mathbf{b})$ evolves along the curve. This evolution is encoded in two scalar functions of arclength: the curvature $\kappa(s)$, which measures how the curve bends in the osculating plane, and the torsion $\tau(s)$, which measures how the curve twists out of that plane.

5.3 Curvature of surfaces*

The geometry of surfaces can be studied through curves that lie on them. At a point P on a surface S with unit normal $\mathbf{n}$, consider any plane containing $\mathbf{n}$. This plane intersects S in a curve passing through P , and this curve has some curvature κ at P . Different choices of plane yield curves with different curvatures.

Definition (Principal curvatures). The *principal curvatures* of a surface S at a point P are the minimum and maximum curvatures among all curves obtained by intersecting S with planes containing the normal at P . These are denoted $\kappa_{\min}$ and $\kappa_{\max}$.

Definition (Gaussian curvature). The *Gaussian curvature* of a surface at a point P is the product of the principal curvatures:

$$K = \kappa_{\min}\kappa_{\max}.$$

The following remarkable theorem, due to Gauss, reveals that Gaussian curvature has a special property.

Theorem (Theorema Egregium). The Gaussian curvature K is *intrinsic* to the surface S : it can be expressed entirely in terms of lengths and angles measured on the surface itself, without reference to the ambient space in which S is embedded.

Remark. This theorem is the foundation of *intrinsic geometry*. It implies that inhabitants of a surface can determine its Gaussian curvature purely through measurements made within the surface, without knowing how (or whether) the surface is embedded in a higher-dimensional space.

The Gaussian curvature controls deviations from Euclidean geometry on the surface. A fundamental example is the angle sum of a triangle.

Definition (Geodesic). A *geodesic* on a surface is a curve of shortest length between two points on the surface.

Consider a geodesic triangle D on a surface S , formed by three geodesics. Let $\theta_1, \theta_2, \theta_3$ be the interior angles of the triangle, measured using scalar products of tangent vectors.

Theorem (Gauss–Bonnet theorem). For a geodesic triangle D on a surface with Gaussian curvature K ,

$$\theta_1 + \theta_2 + \theta_3 = \pi + \int_D K \, dA.$$

Remark. In Euclidean geometry ($K = 0$), the angles of a triangle sum to π . On a sphere ($K > 0$), the sum exceeds π ; on a saddle surface ($K < 0$), the sum is less than π .

6 Div, Grad, Curl and ∇

6.1 The nabla operator

In Cartesian coordinates with orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, we define the *nabla* (or *del*) operator as the formal vector

$$\nabla = \mathbf{e}_i \frac{\partial}{\partial x_i} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right).$$

We assume the basis is right-handed, i.e., $\mathbf{e}_i \times \mathbf{e}_j = \varepsilon_{ijk} \mathbf{e}_k$.

The gradient of a scalar field f is then ∇f , with components $(\nabla f)_i = \frac{\partial f}{\partial x_i}$. We can also apply ∇ to a vector field $\mathbf{F}(\mathbf{r}) = F_i(\mathbf{r})\mathbf{e}_i$ using the scalar or vector product, giving rise to the divergence and curl.

Definition (Divergence). The *divergence* or *div* of $\mathbf{F}$ is

$$\nabla \cdot \mathbf{F} = \frac{\partial F_i}{\partial x_i} = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} + \frac{\partial F_3}{\partial x_3}.$$

Definition (Curl). The *curl* of $\mathbf{F}$ is

$$\nabla \times \mathbf{F} = \varepsilon_{ijk} \frac{\partial F_k}{\partial x_j} \mathbf{e}_i = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

Example (Divergence and curl computation). Let $\mathbf{F} = (xe^z, y^2 \sin x, xyz)$. Then

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x} xe^z + \frac{\partial}{\partial y} y^2 \sin x + \frac{\partial}{\partial z} xyz = e^z + 2y \sin x + xy.$$

and

$$\begin{aligned} \nabla \times \mathbf{F} &= \hat{\mathbf{i}} \left[\frac{\partial}{\partial y} (xyz) - \frac{\partial}{\partial z} (y^2 \sin x) \right] \\ &\quad + \hat{\mathbf{j}} \left[\frac{\partial}{\partial z} (xe^z) - \frac{\partial}{\partial x} (xyz) \right] \\ &\quad + \hat{\mathbf{k}} \left[\frac{\partial}{\partial x} (y^2 \sin x) - \frac{\partial}{\partial y} (xe^z) \right] \\ &= (xz, xe^z - yz, y^2 \cos x). \end{aligned}$$

Remark (Operator ordering). Since ∇ is an operator, ordering matters. For example,

$$\mathbf{F} \cdot \nabla = F_i \frac{\partial}{\partial x_i}$$

is a scalar differential operator, while

$$\mathbf{F} \times \nabla = \mathbf{e}_k \varepsilon_{ijk} F_i \frac{\partial}{\partial x_j}$$

is a vector differential operator. These are distinct from $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$.

Proposition (Linearity of differential operators). Let f, g be scalar fields, $\mathbf{F}, \mathbf{G}$ be vector fields, and λ, μ be constants. Then

$$\begin{aligned}\nabla(\lambda f + \mu g) &= \lambda \nabla f + \mu \nabla g, \\ \nabla \cdot (\lambda \mathbf{F} + \mu \mathbf{G}) &= \lambda \nabla \cdot \mathbf{F} + \mu \nabla \cdot \mathbf{G}, \\ \nabla \times (\lambda \mathbf{F} + \mu \mathbf{G}) &= \lambda \nabla \times \mathbf{F} + \mu \nabla \times \mathbf{G}.\end{aligned}$$

Remark (Dimension dependence). The gradient and divergence can be defined analogously in any dimension n , but the curl is specific to $n = 3$ because it uses the vector (cross) product.

Example (Gradient and divergence of r^α). Consider r^α with $r = |\mathbf{r}|$. We know that $\mathbf{r} = x_i \mathbf{e}_i$. So $r^2 = x_i x_i$. Therefore

$$2r \frac{\partial r}{\partial x_j} = 2x_j,$$

or

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}.$$

So

$$\nabla r^\alpha = \mathbf{e}_i \frac{\partial}{\partial x_i} (r^\alpha) = \mathbf{e}_i \alpha r^{\alpha-1} \frac{\partial r}{\partial x_i} = \alpha r^{\alpha-2} \mathbf{r}.$$

Also,

$$\nabla \cdot \mathbf{r} = \frac{\partial x_i}{\partial x_i} = 3.$$

and

$$\nabla \times \mathbf{r} = \mathbf{e}_k \varepsilon_{ijk} \frac{\partial x_j}{\partial x_i} = 0.$$

Proposition (Leibniz rules for vector operators). Let f, g be scalar fields

and $\mathbf{F}, \mathbf{G}$ be vector fields. Then

$$\begin{aligned}\nabla(fg) &= (\nabla f)g + f(\nabla g), \\ \nabla \cdot (f\mathbf{F}) &= (\nabla f) \cdot \mathbf{F} + f(\nabla \cdot \mathbf{F}), \\ \nabla \times (f\mathbf{F}) &= (\nabla f) \times \mathbf{F} + f(\nabla \times \mathbf{F}), \\ \nabla(\mathbf{F} \cdot \mathbf{G}) &= \mathbf{F} \times (\nabla \times \mathbf{G}) + \mathbf{G} \times (\nabla \times \mathbf{F}) + (\mathbf{F} \cdot \nabla)\mathbf{G} + (\mathbf{G} \cdot \nabla)\mathbf{F}, \\ \nabla \times (\mathbf{F} \times \mathbf{G}) &= \mathbf{F}(\nabla \cdot \mathbf{G}) - \mathbf{G}(\nabla \cdot \mathbf{F}) + (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G}, \\ \nabla \cdot (\mathbf{F} \times \mathbf{G}) &= (\nabla \times \mathbf{F}) \cdot \mathbf{G} - \mathbf{F} \cdot (\nabla \times \mathbf{G}).\end{aligned}$$

Proof. Each identity can be verified by direct computation using suffix notation and the summation convention. $\square$

Remark. The first three identities are straightforward and worth remembering. The last three are more complex; in practice, they can be derived as needed using suffix notation.

Example (Leibniz rules applied to $r^\alpha \mathbf{r}$).

$$\begin{aligned}\nabla \cdot (r^\alpha \mathbf{r}) &= (\nabla r^\alpha) \cdot \mathbf{r} + r^\alpha \nabla \cdot \mathbf{r} \\ &= (\alpha r^{\alpha-2} \mathbf{r}) \cdot \mathbf{r} + r^\alpha (3) \\ &= (\alpha + 3)r^\alpha \\ \nabla \times (r^\alpha \mathbf{r}) &= (\nabla(r^\alpha)) \times \mathbf{r} + r^\alpha (\nabla \times \mathbf{r}) \\ &= \alpha r^{\alpha-2} \mathbf{r} \times \mathbf{r} \\ &= \mathbf{0}\end{aligned}$$

6.2 Second-order derivatives

Two fundamental identities relate the differential operators:

Proposition (Curl-grad and div-curl identities). For any smooth scalar field f and vector field $\mathbf{F}$,

$$\begin{aligned}\nabla \times (\nabla f) &= \mathbf{0}, \\ \nabla \cdot (\nabla \times \mathbf{F}) &= 0.\end{aligned}$$

Proof. Both identities follow from the symmetry of mixed partial derivatives. In suffix notation, the key observation is that

$$\varepsilon_{ijk} \frac{\partial^2 f}{\partial x_i \partial x_j} = 0,$$

since ε_{ijk} is antisymmetric in i, j while $\frac{\partial^2 f}{\partial x_i \partial x_j}$ is symmetric. For example, with $k = 3$,

$$\varepsilon_{ij3} \frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_1 \partial x_2} - \frac{\partial^2 f}{\partial x_2 \partial x_1} = 0.$$

□

The converses of these identities also hold, provided the fields are defined on all of $\mathbb{R}^3$.

Proposition (Scalar potential existence). If $\mathbf{F}$ is a smooth vector field defined on all of $\mathbb{R}^3$ with $\nabla \times \mathbf{F} = \mathbf{0}$, then there exists a scalar field f such that $\mathbf{F} = \nabla f$.

Definition (Conservative/irrotational field and scalar potential). A vector field $\mathbf{F}$ satisfying $\mathbf{F} = \nabla f$ for some scalar field f is called *conservative* or *irrotational*. The function f is called the *scalar potential* of $\mathbf{F}$.

Proposition (Vector potential existence). If $\mathbf{H}$ is a smooth vector field defined on all of $\mathbb{R}^3$ with $\nabla \cdot \mathbf{H} = 0$, then there exists a vector field $\mathbf{A}$ such that $\mathbf{H} = \nabla \times \mathbf{A}$.

Definition (Solenoidal field and vector potential). A vector field $\mathbf{H}$ satisfying $\mathbf{H} = \nabla \times \mathbf{A}$ for some vector field $\mathbf{A}$ is called *solenoidal*. The field $\mathbf{A}$ is called the *vector potential* of $\mathbf{H}$.

Remark. The existence of scalar and vector potentials requires that the field be defined on all of $\mathbb{R}^3$ (or more generally, on a simply connected domain). For fields defined on domains with “holes,” the converse statements may fail.

Definition (Laplacian operator). The *Laplacian operator* is defined by

$$\nabla^2 = \nabla \cdot \nabla = \frac{\partial^2}{\partial x_i \partial x_i} = \left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2} + \frac{\partial^2}{\partial x_3^2} \right).$$

This operation is defined on both scalar and vector fields — on a scalar field,

$$\nabla^2 f = \nabla \cdot (\nabla f),$$

whereas on a vector field,

$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A}).$$

7 Integral theorems

7.1 Statement and examples

There are three fundamental integral theorems in vector calculus: Green's theorem, Stokes' theorem, and the divergence theorem (also known as Gauss' theorem). These are all generalizations of the fundamental theorem of calculus: each states that an n -dimensional integral of a derivative equals an $(n - 1)$ -dimensional integral of the original function over the boundary.

We first state all three theorems with examples. In the following subsection, we show that the three theorems are closely related and prove them by establishing their equivalence.

7.1.1 Green's theorem (in the plane)

Theorem (Green's theorem). Let A be a bounded region in the (x, y) plane with boundary $C = \partial A$, a piecewise smooth, non-self-intersecting closed curve traversed anticlockwise. For smooth functions $P(x, y)$ and $Q(x, y)$,

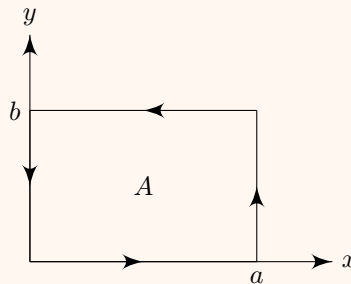
$$\int_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_C P dx + Q dy.$$

Example (Green's theorem, parabola region). Let $Q = xy^2$ and $P = x^2y$. If C is the parabola $y^2 = 4ax$ and the line $x = a$, both with $-2a \leq y \leq 2a$, then Green's theorem says

$$\int_A (y^2 - x^2) dA = \int_C x^2 dx + xy^2 dy.$$

From example sheet 1, each side gives $\frac{104}{105}a^4$.

Example (Green's theorem, rectangle). Let A be a rectangle confined by $0 \leq x \leq a$ and $0 \leq y \leq b$.



Then Green's theorem follows directly from the fundamental theorem of

calculus in 1D. We first consider the first term of Green's theorem:

$$\begin{aligned}\int -\frac{\partial P}{\partial y} \, dA &= \int_0^a \int_0^b -\frac{\partial P}{\partial y} \, dy \, dx \\ &= \int_0^a [-P(x, b) + P(x, 0)] \, dx \\ &= \int_C P \, dx\end{aligned}$$

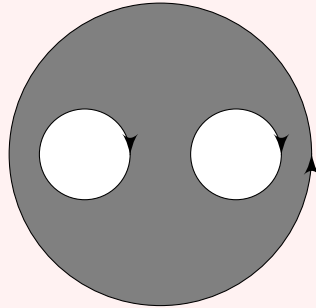
Note that we can convert the 1D integral in the second-to-last line to a line integral around the curve C , since the $P(x, 0)$ and $P(x, b)$ terms give the horizontal part of C , and the lack of dy term means that the integral is nil when integrating the vertical parts.

Similarly,

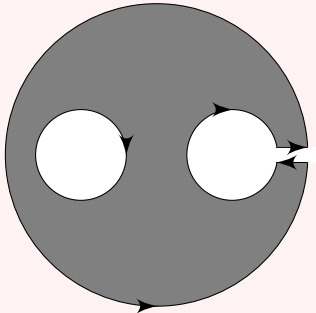
$$\int_A \frac{\partial Q}{\partial x} \, dA = \int_C Q \, dy.$$

Combining them gives Green's theorem.

Remark (Regions with holes). Green's theorem extends to bounded regions A whose boundary ∂A consists of multiple disconnected components (each piecewise smooth, non-self-intersecting, and closed). The exterior boundary is traversed anticlockwise, while interior boundaries (around holes) are traversed clockwise:



This orientation convention can be understood by imagining thin cuts connecting the boundary components, making the region simply connected, and taking the limit as the cuts shrink to zero width:



7.1.2 Stokes' theorem

Theorem (Stokes' theorem). Let S be a smooth, bounded, orientable surface with piecewise smooth boundary ∂S , and let $\mathbf{F}$ be a smooth vector field. Then

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{r}.$$

The orientation of ∂S is determined by the right-hand rule: if the thumb points in the direction of the surface normal $\mathbf{n}$, the fingers curl in the direction of traversal along ∂S .

Remark. Stokes' theorem also holds when ∂S consists of multiple disconnected piecewise smooth closed curves, with orientations determined as in Green's theorem.

Example (Stokes' theorem, spherical cap). Let S be the section of a sphere of radius a with $0 \leq \theta \leq \alpha$. In spherical coordinates,

$$d\mathbf{S} = a^2 \sin \theta \mathbf{e}_r \, d\theta \, d\varphi.$$

Let $\mathbf{F} = (0, xz, 0)$. Then $\nabla \times \mathbf{F} = (-x, 0, z)$. We have previously shown that

$$\int_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = \pi a^3 \cos \alpha \sin^2 \alpha.$$

Our boundary ∂S is

$$\mathbf{r}(\varphi) = a(\sin \alpha \cos \varphi, \sin \alpha \sin \varphi, \cos \alpha).$$

The right hand side of Stokes' is

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} \underbrace{a \sin \alpha \cos \varphi}_x \underbrace{a \cos \alpha}_z \underbrace{a \sin \alpha \cos \varphi \, d\varphi}_{dy} \\ &= a^3 \sin^2 \alpha \cos \alpha \int_0^{2\pi} \cos^2 \varphi \, d\varphi \\ &= \pi a^3 \sin^2 \alpha \cos \alpha, \end{aligned}$$

which agrees with the surface integral, verifying Stokes' theorem.

7.1.3 Divergence theorem (Gauss' theorem)

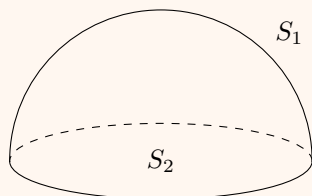
Theorem (Divergence theorem). Let V be a bounded region in $\mathbb{R}^3$ with piecewise smooth boundary ∂V , and let $\mathbf{F}$ be a smooth vector field. Then

$$\int_V \nabla \cdot \mathbf{F} \, dV = \int_{\partial V} \mathbf{F} \cdot d\mathbf{S},$$

where the surface integral is taken with respect to the outward-pointing

normal.

Example (Divergence theorem, hemisphere). Consider a hemisphere.



V is a solid hemisphere

$$x^2 + y^2 + z^2 \leq a^2, \quad z \geq 0,$$

and $\partial V = S_1 + S_2$, the hemisphere and the disc at the bottom.

Take $\mathbf{F} = (0, 0, z + a)$ and $\nabla \cdot \mathbf{F} = 1$. Then

$$\int_V \nabla \cdot \mathbf{F} \, dV = \frac{2}{3}\pi a^3,$$

the volume of the hemisphere.

On S_1 ,

$$d\mathbf{S} = \mathbf{n} \, dS = \frac{1}{a}(x, y, z) \, dS.$$

Then

$$\mathbf{F} \cdot d\mathbf{S} = \frac{1}{a}z(z + a) \, dS = \cos \theta a(\cos \theta + 1) \underbrace{a^2 \sin \theta \, d\theta \, d\varphi}_{dS}.$$

Then

$$\begin{aligned} \int_{S_1} \mathbf{F} \cdot d\mathbf{S} &= a^3 \int_0^{2\pi} d\varphi \int_0^{\pi/2} \sin \theta (\cos^2 \theta + \cos \theta) \, d\theta \\ &= 2\pi a^3 \left[\frac{-1}{3} \cos^3 \theta - \frac{1}{2} \cos^2 \theta \right]_0^{\pi/2} \\ &= \frac{5}{3}\pi a^3. \end{aligned}$$

On S_2 , $d\mathbf{S} = \mathbf{n} \, dS = -(0, 0, 1) \, dS$. Then $\mathbf{F} \cdot d\mathbf{S} = -a \, dS$. So

$$\int_{S_2} \mathbf{F} \cdot d\mathbf{S} = -\pi a^3.$$

So

$$\int_{S_1} \mathbf{F} \cdot d\mathbf{S} + \int_{S_2} \mathbf{F} \cdot d\mathbf{S} = \left(\frac{5}{3} - 1 \right) \pi a^3 = \frac{2}{3}\pi a^3,$$

in accordance with Gauss' theorem.

7.2 Relating and proving integral theorems

We now establish the following equivalences:

- Stokes' theorem $\Leftrightarrow$ Green's theorem
- 2D divergence theorem $\Leftrightarrow$ Green's theorem

We then prove the 2D divergence theorem directly, which establishes all three results. The 3D divergence theorem follows by an analogous argument, though we only sketch the proof as the additional dimension makes the notation more cumbersome.

Proposition (Stokes' theorem implies Green's theorem). Stokes' theorem $\Rightarrow$ Green's theorem.

Proof. Green's theorem concerns regions in $\mathbb{R}^2$, while Stokes' theorem concerns surfaces in $\mathbb{R}^3$. We derive Green's theorem by applying Stokes' theorem to a planar surface.

Let A be a region in the (x, y) plane with boundary $C = \partial A$, parametrised by arc length, $(x(s), y(s), 0)$. Then the tangent to C is

$$\mathbf{t} = \left(\frac{dx}{ds}, \frac{dy}{ds}, 0 \right).$$

Given any $P(x, y)$ and $Q(x, y)$, we can consider the vector field

$$\mathbf{F} = (P, Q, 0),$$

So

$$\nabla \times \mathbf{F} = \left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right).$$

The line integral in Stokes' theorem becomes

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{t} \, ds = \int_C P \, dx + Q \, dy,$$

and the surface integral (with $d\mathbf{S} = \hat{\mathbf{k}} \, dA$ for a surface in the xy -plane) becomes

$$\int_A (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{k}} \, dA = \int_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA. \quad \square$$

Proposition (Green's theorem implies Stokes' theorem). Green's theorem $\Rightarrow$ Stokes' theorem.

Proof. Green's theorem concerns a 2D region, while Stokes' theorem concerns a 3D surface S parametrized by $\mathbf{r}(u, v)$. We apply Green's theorem in the parameter space (u, v) .

Consider a parametrised surface $S = \mathbf{r}(u, v)$ corresponding to the region A in the u, v plane. Write the boundary as $\partial A = (u(t), v(t))$. Then $\partial S = \mathbf{r}(u(t), v(t))$.

We want to prove

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

given

$$\int_{\partial A} F_u \, du + F_v \, dv = \int_A \left(\frac{\partial F_v}{\partial u} - \frac{\partial F_u}{\partial v} \right) \, dA.$$

Doing some pattern-matching, we want

$$\mathbf{F} \cdot d\mathbf{r} = F_u \, du + F_v \, dv$$

for some F_u and F_v .

By the chain rule, we know that

$$d\mathbf{r} = \frac{\partial \mathbf{r}}{\partial u} du + \frac{\partial \mathbf{r}}{\partial v} dv.$$

So we choose

$$F_u = \mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial u}, \quad F_v = \mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial v}.$$

This choice matches the left hand sides of the two equations.

To match the right, recall that

$$(\nabla \times \mathbf{F}) \cdot d\mathbf{S} = (\nabla \times \mathbf{F}) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du \, dv.$$

For the right-hand sides to match, we need

$$\frac{\partial F_v}{\partial u} - \frac{\partial F_u}{\partial v} = (\nabla \times \mathbf{F}) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right). \quad (*)$$

We verify this using suffix notation:

$$\frac{\partial F_v}{\partial u} = \frac{\partial}{\partial u} \left(\mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial v} \right) = \frac{\partial}{\partial u} \left(F_i \frac{\partial x_i}{\partial v} \right) = \left(\frac{\partial F_i}{\partial x_j} \frac{\partial x_j}{\partial u} \right) \frac{\partial x_i}{\partial v} + F_i \frac{\partial x_i}{\partial u \partial v}.$$

Similarly,

$$\frac{\partial F_u}{\partial v} = \frac{\partial}{\partial v} \left(\mathbf{F} \cdot \frac{\partial \mathbf{r}}{\partial u} \right) = \frac{\partial}{\partial v} \left(F_j \frac{\partial x_j}{\partial u} \right) = \left(\frac{\partial F_j}{\partial x_i} \frac{\partial x_i}{\partial v} \right) \frac{\partial x_j}{\partial u} + F_j \frac{\partial x_j}{\partial v \partial u}.$$

So

$$\frac{\partial F_v}{\partial u} - \frac{\partial F_u}{\partial v} = \frac{\partial x_j}{\partial u} \frac{\partial x_i}{\partial v} \left(\frac{\partial F_i}{\partial x_j} - \frac{\partial F_j}{\partial x_i} \right).$$

This is the left hand side of (*).

The right hand side of (*) is

$$\begin{aligned} (\nabla \times \mathbf{F}) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) &= \varepsilon_{ijk} \frac{\partial F_j}{\partial x_i} \varepsilon_{kpq} \frac{\partial x_p}{\partial u} \frac{\partial x_q}{\partial v} \\ &= (\delta_{ip} \delta_{jq} - \delta_{iq} \delta_{jp}) \frac{\partial F_j}{\partial x_i} \frac{\partial x_p}{\partial u} \frac{\partial x_q}{\partial v} \\ &= \left(\frac{\partial F_j}{\partial x_i} - \frac{\partial F_i}{\partial x_j} \right) \frac{\partial x_i}{\partial u} \frac{\partial x_j}{\partial v}. \end{aligned}$$

The two expressions match, so with this choice of F_u and F_v , Green's theorem implies Stokes' theorem. $\square$

Proposition (Green's theorem $\Leftrightarrow$ 2D divergence theorem). Green's theorem and the 2D divergence theorem are equivalent.

Proof. The 2D divergence theorem states that

$$\int_A (\nabla \cdot \mathbf{G}) \, dA = \int_{\partial A} \mathbf{G} \cdot \mathbf{n} \, ds,$$

where $\mathbf{n}$ is the outward-pointing normal.

Given functions P and Q , set $\mathbf{G} = (Q, -P)$. Then

$$\nabla \cdot \mathbf{G} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}.$$

Around the curve $\mathbf{r}(s) = (x(s), y(s))$, $\mathbf{t}(s) = (x'(s), y'(s))$. Then the normal, being perpendicular to $\mathbf{t}$, is $\mathbf{n}(s) = (y'(s), -x'(s))$ (check that it points outwards!). So

$$\mathbf{G} \cdot \mathbf{n} = P \frac{dx}{ds} + Q \frac{dy}{ds}.$$

Then we can expand out the integrals to obtain

$$\int_C \mathbf{G} \cdot \mathbf{n} \, ds = \int_C P \, dx + Q \, dy,$$

and

$$\int_A (\nabla \cdot \mathbf{G}) \, dA = \int_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA.$$

The 2D divergence theorem asserts that the two left-hand sides are equal, while Green's theorem asserts that the two right-hand sides are equal. Thus the two theorems are equivalent. $\square$

Proposition (2D divergence theorem). Let A be a bounded region in $\mathbb{R}^2$ with piecewise smooth boundary $C = \partial A$. For a smooth vector field $\mathbf{G}$,

$$\int_A (\nabla \cdot \mathbf{G}) \, dA = \int_C \mathbf{G} \cdot \mathbf{n} \, ds,$$

where $\mathbf{n}$ is the outward-pointing unit normal.

Proof. By linearity, it suffices to prove the result for vector fields with only a vertical component; the horizontal case is analogous, and the general case follows by superposition.

We also assume that A is a simple convex region. More complicated regions can be decomposed into simple pieces, and the theorem applied to each piece.

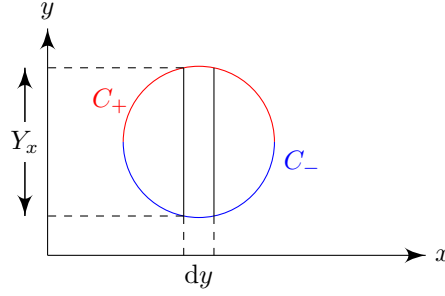
Suppose $\mathbf{G} = G(x, y)\hat{\mathbf{j}}$. Then

$$\nabla \cdot \mathbf{G} = \frac{\partial G}{\partial y}.$$

Then

$$\int_A \nabla \cdot \mathbf{G} \, dA = \int_X \left(\int_{Y_x} \frac{\partial G}{\partial y} \, dy \right) \, dx.$$

Divide the boundary C into upper and lower parts, C_+ and C_- , described by $y = y_+(x)$ and $y = y_-(x)$ respectively:



We see that the boundary of Y_x at any specific x is given by $y_-(x)$ and $y_+(x)$. Hence by the Fundamental theorem of Calculus,

$$\int_{Y_x} \frac{\partial G}{\partial y} dy = \int_{y_-(x)}^{y_+(x)} \frac{\partial G}{\partial y} dy = G(x, y_+(x)) - G(x, y_-(x)).$$

To compute the full area integral, we want to integrate over all x . However, the divergence theorem talks in terms of ds , not dx . So we need to find some way to relate ds and dx . If we move a distance δs , the change in x is $\delta s \cos \theta$, where θ is the angle between the tangent and the horizontal. But θ is also the angle between the normal and the vertical. So $\cos \theta = \mathbf{n} \cdot \hat{\mathbf{j}}$. Therefore $dx = \hat{\mathbf{j}} \cdot \mathbf{n} ds$.

In particular, $G dx = G \hat{\mathbf{j}} \cdot \mathbf{n} ds = \mathbf{G} \cdot \mathbf{n} ds$, since $\mathbf{G} = G \hat{\mathbf{j}}$.

However, at C_- , $\mathbf{n}$ points downwards, so $\mathbf{n} \cdot \hat{\mathbf{j}}$ happens to be negative. So, actually, at C_- , $dx = -\mathbf{G} \cdot \mathbf{n} ds$.

Therefore, our full integral is

$$\begin{aligned} \int_A \nabla \cdot \mathbf{G} dA &= \int_X \left(\int_{y_x} \frac{\partial G}{\partial y} dY \right) dx \\ &= \int_X G(x, y_+(x)) - G(x, y_-(x)) dx \\ &= \int_{C_+} \mathbf{G} \cdot \mathbf{n} ds + \int_{C_-} \mathbf{G} \cdot \mathbf{n} ds \\ &= \int_C \mathbf{G} \cdot \mathbf{n} ds. \end{aligned} \quad \square$$

Remark (Sketch of 3D divergence theorem proof). The 3D divergence theorem is proved analogously. Consider a purely vertical vector field $\mathbf{F} = F(x, y, z) \hat{\mathbf{k}}$. Then

$$\int_V \nabla \cdot \mathbf{F} dV = \int_D \left(\int_{Z_{xy}} \frac{\partial F}{\partial z} dz \right) dA,$$

where D is the projection of V onto the xy -plane and Z_{xy} is the range of z values above each point $(x, y) \in D$. Splitting $S = \partial V$ into top and bottom parts S_+ and S_- (where $\hat{\mathbf{k}} \cdot \mathbf{n} \geq 0$ and $\hat{\mathbf{k}} \cdot \mathbf{n} < 0$ respectively), parametrized by $z = z_+(x, y)$ and $z = z_-(x, y)$, the integral becomes

$$\int_V \nabla \cdot \mathbf{F} dV = \int_D (F(x, y, z_+) - F(x, y, z_-)) dA = \int_S \mathbf{F} \cdot \mathbf{n} dS.$$

The general case follows by decomposing $\mathbf{F}$ into components along each coordinate axis.

8 Some applications of integral theorems

8.1 Integral expressions for div and curl

The integral theorems provide coordinate-free definitions of divergence and curl that clarify their geometric meaning.

Applying the divergence theorem to a small volume V containing a point $\mathbf{r}_0$,

$$\int_{\partial V} \mathbf{F} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{F} \, dV \approx (\nabla \cdot \mathbf{F})(\mathbf{r}_0) \text{vol}(V).$$

Taking the limit as V shrinks to $\mathbf{r}_0$ yields:

Proposition (Integral definition of divergence).

$$(\nabla \cdot \mathbf{F})(\mathbf{r}_0) = \lim_{\text{diam}(V) \rightarrow 0} \frac{1}{\text{vol}(V)} \int_{\partial V} \mathbf{F} \cdot d\mathbf{S},$$

where the limit is taken over volumes V containing $\mathbf{r}_0$.

Similarly, applying Stokes' theorem to a small surface A containing $\mathbf{r}_0$ with unit normal $\mathbf{n}$,

$$\int_{\partial A} \mathbf{F} \cdot d\mathbf{r} = \int_A (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dA \approx \mathbf{n} \cdot (\nabla \times \mathbf{F})(\mathbf{r}_0) \text{area}(A).$$

Proposition (Integral definition of curl).

$$\mathbf{n} \cdot (\nabla \times \mathbf{F})(\mathbf{r}_0) = \lim_{\text{diam}(A) \rightarrow 0} \frac{1}{\text{area}(A)} \int_{\partial A} \mathbf{F} \cdot d\mathbf{r},$$

where the limit is taken over surfaces A containing $\mathbf{r}_0$ with normal $\mathbf{n}$.

Remark. These definitions are independent of the choice of coordinate system and provide the basis for physical interpretations of divergence and curl.

Example (Fluid flow interpretation of divergence and curl). Let $\mathbf{u}$ be the velocity field of a fluid flow. The flux integral $\int_S \mathbf{u} \cdot d\mathbf{S}$ gives the rate at which fluid crosses the surface S .

Divergence as expansion rate: Consider a small volume V of fluid. The rate of change of its volume is

$$\dot{V} = \int_{\partial V} \mathbf{u} \cdot d\mathbf{S}.$$

By the integral definition of divergence,

$$\nabla \cdot \mathbf{u} = \lim_{V \rightarrow 0} \frac{\dot{V}}{V},$$

which is the relative rate of volume expansion. For example, if $\mathbf{u}(\mathbf{r}) = \alpha \mathbf{r}$ (radial outflow from the origin), then $\nabla \cdot \mathbf{u} = 3\alpha$, indicating uniform expansion throughout space.

Curl as rotation rate: Consider a disc A of radius a with unit normal $\mathbf{n}$. The circulation around its boundary is

$$\int_{\partial A} \mathbf{u} \cdot d\mathbf{r} = \int_{\partial A} \mathbf{u} \cdot \mathbf{t} \, ds = 2\pi a \times (\text{average tangential velocity}).$$

Define the local angular velocity as

$$\omega = \frac{1}{a} \times (\text{average of } \mathbf{u} \cdot \mathbf{t}).$$

As $a \rightarrow 0$, the average tangential velocity decreases (a smooth field varies less on smaller scales), but the ratio ω approaches a finite limit. Since $\int_{\partial A} \mathbf{u} \cdot d\mathbf{r} = 2\pi a^2 \omega$, the integral definition gives

$$\mathbf{n} \cdot (\nabla \times \mathbf{u}) = \lim_{a \rightarrow 0} \frac{1}{\pi a^2} \int_{\partial A} \mathbf{u} \cdot d\mathbf{r} = 2\omega.$$

Thus the curl measures twice the local angular velocity of the fluid. For rigid body rotation with angular velocity $\boldsymbol{\omega}$, the velocity field is $\mathbf{u} = \boldsymbol{\omega} \times \mathbf{r}$, and

$$\nabla \times (\boldsymbol{\omega} \times \mathbf{r}) = 2\boldsymbol{\omega}.$$

8.2 Conservative fields and scalar potentials

Definition (Conservative field). A vector field $\mathbf{F}$ is *conservative* if any of the following equivalent conditions holds:

- (i) $\mathbf{F} = \nabla f$ for some scalar field f (called the *scalar potential*);
- (ii) $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on the endpoints of C , not on the path;
- (iii) $\nabla \times \mathbf{F} = \mathbf{0}$.

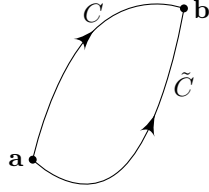
We now prove these conditions are equivalent. The implications (i) $\Rightarrow$ (ii) and (i) $\Rightarrow$ (iii) were established earlier:

$$(i) \Rightarrow (ii): \quad \int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a}), \quad (i) \Rightarrow (iii): \quad \nabla \times (\nabla f) = \mathbf{0}.$$

It remains to show (iii) $\Rightarrow$ (ii) and (ii) $\Rightarrow$ (i).

Proposition (Path independence, sufficient condition). If (iii) $\nabla \times \mathbf{F} = \mathbf{0}$, then (ii) $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of C .

Proof. Given $\mathbf{F}(\mathbf{r})$ satisfying $\nabla \times \mathbf{F} = \mathbf{0}$, let C and $\tilde{C}$ be any two curves from $\mathbf{a}$ to $\mathbf{b}$.



If S is any surface with boundary $\partial S = C - \tilde{C}$, By Stokes' theorem,

$$\int_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot d\mathbf{r} - \int_{\tilde{C}} \mathbf{F} \cdot d\mathbf{r}.$$

But $\nabla \times \mathbf{F} = 0$. So

$$\int_C \mathbf{F} \cdot d\mathbf{r} - \int_{\tilde{C}} \mathbf{F} \cdot d\mathbf{r} = 0,$$

or

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{\tilde{C}} \mathbf{F} \cdot d\mathbf{r}. \quad \square$$

Proposition (Path independence, necessary condition). If (ii) $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of C for fixed end points and orientation, then (i) $\mathbf{F} = \nabla f$ for some scalar field f .

Proof. We fix $\mathbf{a}$ and define $f(\mathbf{r}) = \int_C \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}'$ for any curve from $\mathbf{a}$ to $\mathbf{r}$. Assuming (ii), f is well-defined. For small changes $\mathbf{r}$ to $\mathbf{r} + \delta\mathbf{r}$, there is a small extension of C by δC . Then

$$\begin{aligned} f(\mathbf{r} + \delta\mathbf{r}) &= \int_{C+\delta C} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}' \\ &= \int_C \mathbf{F} \cdot d\mathbf{r}' + \int_{\delta C} \mathbf{F} \cdot d\mathbf{r}' \\ &= f(\mathbf{r}) + \mathbf{F}(\mathbf{r}) \cdot \delta\mathbf{r} + o(\delta\mathbf{r}). \end{aligned}$$

So

$$\delta f = f(\mathbf{r} + \delta\mathbf{r}) - f(\mathbf{r}) = \mathbf{F}(\mathbf{r}) \cdot \delta\mathbf{r} + o(\delta\mathbf{r}).$$

But the definition of grad is exactly

$$\delta f = \nabla f \cdot \delta\mathbf{r} + o(\delta\mathbf{r}).$$

So we have $\mathbf{F} = \nabla f$. $\square$

Remark (Simply connected domains). The above results assume $\mathbf{F}$ is defined on all of $\mathbb{R}^3$, but they also hold on any *simply connected* domain D . A domain is simply connected if any closed loop can be continuously shrunk to a point, or equivalently, if any two curves with the same endpoints can be smoothly deformed into one another.

The proof that (iii) $\Rightarrow$ (ii) requires the existence of a surface S bounded by $C - \tilde{C}$. Such a surface exists precisely when C and $\tilde{C}$ can be deformed into each other.

If D is not simply connected, a scalar potential f may be multi-valued. However, one can often restrict to a simply connected subset $D_0 \subseteq D$ on which f is single-valued.

Example (Multi-valued potential on a non-simply-connected domain). Consider the vector field

$$\mathbf{F} = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2}, 0 \right),$$

defined on $D = \mathbb{R}^3 \setminus \{z\text{-axis}\}$, which is not simply connected. One can verify that $\nabla \times \mathbf{F} = \mathbf{0}$ on D . Formally, $\mathbf{F} = \nabla f$ where

$$f = \arctan \frac{y}{x},$$

but this potential is multi-valued (the arctangent has multiple branches). The non-trivial topology manifests when we integrate around the unit circle $x^2 + y^2 = 1$ in the $z = 0$ plane:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi \neq 0.$$

For a truly conservative field, all closed loop integrals would vanish. To obtain a single-valued potential, we can restrict to the simply connected domain

$$D_0 = \mathbb{R}^3 \setminus \{(x, y, z) : x \leq 0, y = 0\},$$

obtained by removing a half-plane. On D_0 , no closed curve can encircle the z -axis, and the potential becomes single-valued.

8.3 Conservation laws

Definition (Continuity equation). Let $\rho(\mathbf{r}, t)$ be the density (amount per unit volume) of some conserved quantity, and let $\mathbf{j}(\mathbf{r}, t)$ be the corresponding flux density (flow rate per unit area). The *continuity equation* (or *conservation equation*) is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0.$$

This equation expresses *local* conservation: the quantity cannot appear or disappear, but must flow continuously from place to place.

Proposition (Integral form of conservation). Let V be a fixed region with boundary $S = \partial V$, and let $Q(t) = \int_V \rho(\mathbf{r}, t) dV$ be the total amount of the

conserved quantity in V . Then

$$\frac{dQ}{dt} = - \int_S \mathbf{j} \cdot d\mathbf{S}.$$

Proof. By the continuity equation and the divergence theorem,

$$\frac{dQ}{dt} = \int_V \frac{\partial \rho}{\partial t} dV = - \int_V \nabla \cdot \mathbf{j} dV = - \int_S \mathbf{j} \cdot d\mathbf{S}. \quad \square$$

This states that the rate of change of Q in V equals the net inward flux through the boundary: the quantity cannot disappear from V except by flowing out through ∂V .

Remark (Global conservation). If $\mathbf{j} \rightarrow \mathbf{0}$ sufficiently rapidly as $|\mathbf{r}| \rightarrow \infty$, then taking V to be all of $\mathbb{R}^3$ (as a limit of large spheres) gives $\frac{dQ_{\text{total}}}{dt} = 0$: the total amount of the quantity in the universe is constant.

Example (Charge conservation). Let $\rho(\mathbf{r}, t)$ be the electric charge density and $\mathbf{j}(\mathbf{r}, t)$ be the current density. The continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0$$

expresses conservation of electric charge: charge can only change in a region by flowing in or out as current.

Example (Mass conservation and incompressibility). For a fluid with mass density $\rho(\mathbf{r}, t)$ and velocity field $\mathbf{u}(\mathbf{r}, t)$, the mass flux is $\mathbf{j} = \rho\mathbf{u}$. The continuity equation becomes

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho\mathbf{u}) = 0,$$

which is the equation of mass conservation for a fluid.

For an *incompressible* fluid, the density ρ is constant (independent of $\mathbf{r}$ and t). The continuity equation then reduces to

$$\nabla \cdot \mathbf{u} = 0,$$

stating that the velocity field is divergence-free.

9 Orthogonal curvilinear coordinates

9.1 Line, area and volume elements

A coordinate system provides a way to specify points in space using a set of numbers. In three dimensions, we describe a coordinate system by a smooth, invertible function $\mathbf{r}(u, v, w)$ mapping coordinate values to position vectors. By the chain rule,

$$d\mathbf{r} = \frac{\partial \mathbf{r}}{\partial u} du + \frac{\partial \mathbf{r}}{\partial v} dv + \frac{\partial \mathbf{r}}{\partial w} dw.$$

For the coordinates to be non-degenerate, the tangent vectors $\frac{\partial \mathbf{r}}{\partial u}$, $\frac{\partial \mathbf{r}}{\partial v}$, $\frac{\partial \mathbf{r}}{\partial w}$ must be linearly independent at each point, i.e.,

$$\frac{\partial \mathbf{r}}{\partial u} \cdot \left(\frac{\partial \mathbf{r}}{\partial v} \times \frac{\partial \mathbf{r}}{\partial w} \right) \neq 0.$$

These tangent vectors point along the coordinate curves (curves where two coordinates are held fixed).

Definition (Orthogonal curvilinear coordinates). Coordinates (u, v, w) are *orthogonal curvilinear coordinates* if the tangent vectors $\frac{\partial \mathbf{r}}{\partial u}$, $\frac{\partial \mathbf{r}}{\partial v}$, $\frac{\partial \mathbf{r}}{\partial w}$ are mutually orthogonal at every point.

Definition (Scale factors). For orthogonal curvilinear coordinates, the *scale factors* h_u , h_v , h_w are defined by

$$h_u = \left| \frac{\partial \mathbf{r}}{\partial u} \right|, \quad h_v = \left| \frac{\partial \mathbf{r}}{\partial v} \right|, \quad h_w = \left| \frac{\partial \mathbf{r}}{\partial w} \right|.$$

The corresponding unit vectors are

$$\mathbf{e}_u = \frac{1}{h_u} \frac{\partial \mathbf{r}}{\partial u}, \quad \mathbf{e}_v = \frac{1}{h_v} \frac{\partial \mathbf{r}}{\partial v}, \quad \mathbf{e}_w = \frac{1}{h_w} \frac{\partial \mathbf{r}}{\partial w},$$

which form an orthonormal right-handed basis (so $\mathbf{e}_u \times \mathbf{e}_v = \mathbf{e}_w$).

The line element can then be written as

$$d\mathbf{r} = h_u \mathbf{e}_u du + h_v \mathbf{e}_v dv + h_w \mathbf{e}_w dw.$$

Example (Scale factors in standard coordinates).

(i) In cartesian coordinates, $\mathbf{r}(x, y, z) = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$. Then $h_x = h_y = h_z = 1$, and $\mathbf{e}_x = \hat{\mathbf{i}}$, $\mathbf{e}_y = \hat{\mathbf{j}}$ and $\mathbf{e}_z = \hat{\mathbf{k}}$.

(ii) In cylindrical polars, $\mathbf{r}(\rho, \varphi, z) = \rho[\cos \varphi \hat{\mathbf{i}} + \sin \varphi \hat{\mathbf{j}}] + z\hat{\mathbf{k}}$. Then $h_\rho = h_z = 1$, and

$$h_\varphi = \left| \frac{\partial \mathbf{r}}{\partial \varphi} \right| = |(-\rho \sin \varphi, \rho \cos \varphi, 0)| = \rho.$$

The basis vectors $\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{e}_z$ are as in section 1.

(iii) In spherical polars,

$$\mathbf{r}(r, \theta, \varphi) = r(\cos \varphi \sin \theta \hat{\mathbf{i}} + \sin \varphi \sin \theta \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}}).$$

Then $h_r = 1, h_\theta = r$ and $h_\varphi = r \sin \theta$.

Proposition (Area and volume elements in curvilinear coordinates). For a coordinate surface with w constant (parametrized by u and v), the vector area element is

$$d\mathbf{S} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} du dv = h_u h_v \mathbf{e}_w du dv.$$

The volume element is

$$dV = \frac{\partial \mathbf{r}}{\partial u} \cdot \left(\frac{\partial \mathbf{r}}{\partial v} \times \frac{\partial \mathbf{r}}{\partial w} \right) du dv dw = h_u h_v h_w du dv dw.$$

Remark. Geometrically, an infinitesimal coordinate box has edges of lengths $h_u \delta u$, $h_v \delta v$, and $h_w \delta w$ along the three orthogonal directions.

9.2 Gradient, divergence, and curl

We now derive expressions for the differential operators in orthogonal curvilinear coordinates.

For a scalar field $f(\mathbf{r}(u, v, w))$, the chain rule gives

$$df = \frac{\partial f}{\partial u} du + \frac{\partial f}{\partial v} dv + \frac{\partial f}{\partial w} dw.$$

Comparing with $df = (\nabla f) \cdot d\mathbf{r}$ and using the expression for $d\mathbf{r}$, we obtain:

Proposition (Gradient in curvilinear coordinates).

$$\nabla f = \frac{1}{h_u} \frac{\partial f}{\partial u} \mathbf{e}_u + \frac{1}{h_v} \frac{\partial f}{\partial v} \mathbf{e}_v + \frac{1}{h_w} \frac{\partial f}{\partial w} \mathbf{e}_w.$$

Example (Gradient in spherical polars). Take $f = r \sin \theta \cos \varphi$ in spherical polars. Then

$$\begin{aligned} \nabla f &= \sin \theta \cos \varphi \mathbf{e}_r + \frac{1}{r} (r \cos \theta \cos \varphi) \mathbf{e}_\theta + \frac{1}{r \sin \theta} (-r \sin \theta \sin \varphi) \mathbf{e}_\varphi \\ &= \cos \varphi (\sin \theta \mathbf{e}_r + \cos \theta \mathbf{e}_\theta) - \sin \varphi \mathbf{e}_\varphi. \end{aligned}$$

The nabla operator in curvilinear coordinates is thus

$$\nabla = \frac{1}{h_u} \mathbf{e}_u \frac{\partial}{\partial u} + \frac{1}{h_v} \mathbf{e}_v \frac{\partial}{\partial v} + \frac{1}{h_w} \mathbf{e}_w \frac{\partial}{\partial w}.$$

For a vector field $\mathbf{F} = F_u \mathbf{e}_u + F_v \mathbf{e}_v + F_w \mathbf{e}_w$, the divergence and curl take the following forms.

Proposition (Divergence in curvilinear coordinates).

$$\nabla \cdot \mathbf{F} = \frac{1}{h_u h_v h_w} \left[\frac{\partial}{\partial u} (h_v h_w F_u) + \frac{\partial}{\partial v} (h_u h_w F_v) + \frac{\partial}{\partial w} (h_u h_v F_w) \right].$$

Proposition (Curl in curvilinear coordinates).

$$\nabla \times \mathbf{F} = \frac{1}{h_u h_v h_w} \begin{vmatrix} h_u \mathbf{e}_u & h_v \mathbf{e}_v & h_w \mathbf{e}_w \\ \frac{\partial}{\partial u} & \frac{\partial}{\partial v} & \frac{\partial}{\partial w} \\ h_u F_u & h_v F_v & h_w F_w \end{vmatrix}.$$

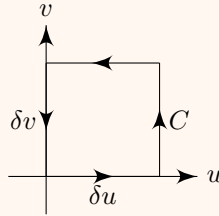
Remark (Derivation methods). There are several approaches to deriving these formulas:

- (i) Apply $\nabla \cdot$ or $\nabla \times$ directly, differentiating the basis vectors $\mathbf{e}_u$, $\mathbf{e}_v$, $\mathbf{e}_w$ explicitly (these depend on position).
- (ii) Express $\mathbf{F}$ in terms of ∇u , ∇v , ∇w and use the identities $\nabla \times (\nabla f) = \mathbf{0}$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$.
- (iii) Use the integral definitions of divergence and curl.

Example (Derivation of curl using integral definition). We illustrate the third approach. Recall that

$$\mathbf{n} \cdot (\nabla \times \mathbf{F}) = \lim_{A \rightarrow 0} \frac{1}{A} \int_{\partial A} \mathbf{F} \cdot d\mathbf{r}.$$

To find the $\mathbf{e}_w$ component, consider a small coordinate rectangle in the surface $w = \text{const}$, with sides δu and δv :



This rectangle has area $h_u h_v \delta u \delta v$ and normal $\mathbf{e}_w$. Integrating around

the boundary C and using linear approximations,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &\approx F_u(u, v) h_u(u, v) \delta u + F_v(u + \delta u, v) h_v(u + \delta u, v) \delta v \\ &\quad - F_u(u, v + \delta v) h_u(u, v + \delta v) \delta u - F_v(u, v) h_v(u, v) \delta v \\ &\approx \left[\frac{\partial}{\partial u} (h_v F_v) - \frac{\partial}{\partial v} (h_u F_u) \right] \delta u \delta v. \end{aligned}$$

Dividing by the area and taking the limit gives

$$\mathbf{e}_w \cdot (\nabla \times \mathbf{F}) = \frac{1}{h_u h_v} \left[\frac{\partial}{\partial u} (h_v F_v) - \frac{\partial}{\partial v} (h_u F_u) \right].$$

The other components follow by cyclic permutation. The divergence formula can be derived similarly.

Example (Curl in spherical polars). Let $\mathbf{A} = \frac{1}{r} \tan \frac{\theta}{2} \mathbf{e}_\varphi$ in spherical polars. Then

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & r \sin \theta \mathbf{e}_\varphi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ 0 & 0 & r \sin \theta \cdot \frac{1}{r} \tan \frac{\theta}{2} \end{vmatrix} = \frac{\mathbf{e}_r}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta \tan \frac{\theta}{2} \right] = \frac{1}{r^2} \mathbf{e}_r.$$

10 Gauss' Law and Poisson's equation

10.1 Laws of gravitation

Consider a distribution of mass producing a gravitational force $\mathbf{F}$ on a point mass m at $\mathbf{r}$. The total force is a sum of contributions from each part of the mass distribution, and is proportional to m . Write

$$\mathbf{F} = m\mathbf{g}(\mathbf{r}),$$

Definition (Gravitational field). $\mathbf{g}(\mathbf{r})$ is the *gravitational field, acceleration due to gravity, or force per unit mass*.

The gravitational field is conservative, i.e.,

$$\oint_C \mathbf{g} \cdot d\mathbf{r} = 0$$

for any closed curve C . Physically, this means the work done by gravity around any closed path is zero, reflecting conservation of gravitational potential energy.

The gravitational field satisfies the following fundamental law:

Law (Gauss' law for gravitation). Given any volume V bounded by closed surface S ,

$$\int_S \mathbf{g} \cdot d\mathbf{S} = -4\pi GM,$$

where G is Newton's gravitational constant, and M is the total mass contained in V .

These equations determine $\mathbf{g}(\mathbf{r})$ from a mass distribution.

Example (Newton's law from Gauss' law). We can obtain Newton's law of gravitation from Gauss' law together with an assumption about symmetry.

Consider a total mass M distributed with a spherical symmetry about the origin $\mathbf{O}$, with all the mass contained within some radius $r = a$. By spherical symmetry, we have $\mathbf{g}(\mathbf{r}) = g(\mathbf{r})\hat{\mathbf{r}}$.

Consider Gauss' law with S being a sphere of radius $r = R > a$. Then $\hat{\mathbf{n}} = \hat{\mathbf{r}}$. So

$$\int_S \mathbf{g} \cdot d\mathbf{S} = \int_S g(R)\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} dS = \int_S g(R) dS = 4\pi R^2 g(R).$$

By Gauss' law, we obtain

$$4\pi R^2 g(R) = -4\pi GM.$$

So

$$g(R) = -\frac{GM}{R^2}$$

for $R > a$.

Therefore the gravitational force on a mass m at $\mathbf{r}$ is

$$\mathbf{F}(\mathbf{r}) = -\frac{GMm}{r^2}\hat{\mathbf{r}}.$$

If we take the limit as $a \rightarrow 0$, we get a point mass M at the origin. Then we recover Newton's law of gravitation for point masses.

Proposition. The gravitational field is irrotational: $\nabla \times \mathbf{g} = \mathbf{0}$.

Proof. Since $\oint_C \mathbf{g} \cdot d\mathbf{r} = 0$ for any closed curve C , Stokes' theorem gives

$$\int_S \nabla \times \mathbf{g} \cdot d\mathbf{S} = 0$$

for any surface S bounded by C . Since this holds for arbitrary surfaces, we conclude $\nabla \times \mathbf{g} = \mathbf{0}$. $\square$

Remark. While Gauss' law in integral form is useful when the mass distribution has sufficient symmetry, it can be difficult to apply in general. For arbitrary distributions, the differential form is often more convenient.

To derive the differential form, suppose

$$M = \int_V \rho(\mathbf{r}) dV,$$

where ρ is the mass density. Then by Gauss' theorem

$$\int_S \mathbf{g} \cdot d\mathbf{S} = -4\pi GM \Rightarrow \int_V \nabla \cdot \mathbf{g} dV = \int_V -4\pi G\rho dV.$$

Since this is true for all V , we must have

Law (Gauss' Law for gravitation in differential form).

$$\nabla \cdot \mathbf{g} = -4\pi G\rho.$$

Definition (Gravitational potential). Since $\nabla \times \mathbf{g} = \mathbf{0}$, there exists a scalar field $\varphi(\mathbf{r})$ such that $\mathbf{g} = -\nabla\varphi$. This φ is called the *gravitational potential*.

Substituting into Gauss' law in differential form gives

$$\nabla^2\varphi = 4\pi G\rho.$$

For the spherically symmetric example, we find

$$\varphi(r) = -\frac{GM}{r}$$

for $r > a$.

10.2 Laws of electrostatics

Consider a distribution of electric charge at rest. The charges produce a force on a test charge q at position $\mathbf{r}$, proportional to q .

Definition (Electric field). The *electric field* $\mathbf{E}(\mathbf{r})$ is defined by $\mathbf{F} = q\mathbf{E}(\mathbf{r})$, where $\mathbf{F}$ is the force on a test charge q at $\mathbf{r}$. Equivalently, $\mathbf{E}$ is the force per unit charge.

The electrostatic field is conservative:

$$\oint_C \mathbf{E} \cdot d\mathbf{r} = 0$$

for any closed curve C . It also satisfies

Law (Gauss' law for electrostatic forces).

$$\int_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\varepsilon_0},$$

where ε_0 is the *permittivity of free space*, or *electric constant*.

Applying the divergence theorem as in the gravitational case yields the differential form:

Law (Gauss' law for electrostatic forces in differential form).

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}.$$

Remark. In electrostatics (no time-varying magnetic fields), the electric field is irrotational: $\nabla \times \mathbf{E} = \mathbf{0}$. This follows from the conservative property and Stokes' theorem, just as for the gravitational field.

Since $\nabla \times \mathbf{E} = \mathbf{0}$, we can write $\mathbf{E} = -\nabla\varphi$.

Definition (Electrostatic potential). If we write $\mathbf{E} = -\nabla\varphi$, then φ is the *electrostatic potential*, and

$$\nabla^2\varphi = -\frac{\rho}{\varepsilon_0}.$$

Example (Coulomb's law from Gauss' law). Take a spherically symmetric charge distribution about O with total charge Q . Suppose all charge is contained within a radius $r = a$. Then similar to the gravitational case, we have

$$\mathbf{E}(\mathbf{r}) = \frac{Q\hat{\mathbf{r}}}{4\pi\varepsilon_0 r^2},$$

and

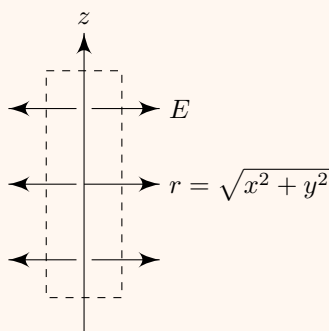
$$\varphi(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0 r}.$$

As $a \rightarrow 0$, we get *point charges*. From $\mathbf{E}$, we can recover Coulomb's law for the force on another charge q at $\mathbf{r}$:

$$\mathbf{F} = q\mathbf{E} = \frac{qQ\hat{\mathbf{r}}}{4\pi\epsilon_0 r^2}.$$

Example (Line charge). Consider an infinite line with uniform charge density *per unit length* σ .

We use cylindrical polar coordinates:



By symmetry, the field is radial, i.e.

$$\mathbf{E}(r) = E(r)\hat{\mathbf{r}}.$$

Pick S to be a cylinder of length L and radius r . We know that the end caps do not contribute to the flux since the field lines are perpendicular to the normal. Also, the curved surface has area $2\pi rL$. Then by Gauss' law in integral form,

$$\int_S \mathbf{E} \cdot d\mathbf{S} = E(r)2\pi rL = \frac{\sigma L}{\epsilon_0}.$$

So

$$\mathbf{E}(r) = \frac{\sigma}{2\pi\epsilon_0 r}\hat{\mathbf{r}}.$$

Note that the field varies as $1/r$, not $1/r^2$. This slower decay compared to a point charge reflects the fact that the line charge extends infinitely in one direction.

10.3 Poisson's Equation and Laplace's equation

Definition (Poisson's equation). The *Poisson's equation* is

$$\nabla^2 \varphi = -\rho,$$

where ρ is given and $\varphi(\mathbf{r})$ is to be solved.

Remark. The gravitational and electrostatic potentials satisfy Poisson's equation with ρ replaced by $-4\pi G\rho_{\text{mass}}$ and $\rho_{\text{charge}}/\varepsilon_0$ respectively.

When $\rho = 0$, we get

Definition (Laplace's equation). Laplace's equation is

$$\nabla^2 \varphi = 0.$$

Example (Irrotational incompressible flow). For irrotational and incompressible fluid flow with velocity field $\mathbf{u}(\mathbf{r})$, irrotationality implies $\mathbf{u} = \nabla \varphi$ for some *velocity potential* φ . Incompressibility gives $\nabla \cdot \mathbf{u} = 0$ (cf. Section ??). Combining these, $\nabla^2 \varphi = 0$.

Remark. Expressions for ∇^2 in non-Cartesian coordinates can be derived from the curvilinear coordinate formulas, but are somewhat complicated in general.

We focus on cases with spherical or cylindrical symmetry. If $\varphi = \varphi(r)$ depends only on the radial coordinate r , then $\nabla \varphi = \varphi'(r)\hat{\mathbf{r}}$ and Laplace's equation reduces to an ordinary differential equation.

Proposition (Laplace's equation with radial symmetry).

(i) For spherical symmetry ($r^2 = x^2 + y^2 + z^2$),

$$\nabla^2 \varphi = \varphi'' + \frac{2}{r}\varphi' = \frac{1}{r^2}(r^2\varphi')' = 0,$$

with general solution $\varphi = \frac{A}{r} + B$.

(ii) For cylindrical symmetry ($r^2 = x^2 + y^2$),

$$\nabla^2 \varphi = \varphi'' + \frac{1}{r}\varphi' = \frac{1}{r}(r\varphi')' = 0,$$

with general solution $\varphi = A \ln r + B$.

Solutions to Poisson's equation with the same symmetries can be obtained by finding particular integrals and adding them to the homogeneous solutions

above.

For example, for a spherically symmetric solution of $\nabla^2\varphi = -\rho_0$, with ρ_0 constant, recall that $\nabla^2 r^\alpha = \alpha(\alpha+1)r^{\alpha-2}$. Taking $\alpha = 2$, we find the particular integral

$$\varphi = -\frac{\rho_0}{6}r^2,$$

So the general solution with spherical symmetry and constant ρ_0 is

$$\varphi(r) = \frac{A}{r} + B - \frac{1}{6}\rho_0 r^2.$$

To determine the constants A and B , we must specify boundary conditions. If φ is defined on all of $\mathbb{R}^3$, we often require $\varphi \rightarrow 0$ as $|\mathbf{r}| \rightarrow \infty$. For φ defined on a bounded volume V , there are two standard types of boundary conditions on ∂V :

Definition (Boundary conditions). Let $\mathbf{n}$ denote the outward unit normal on ∂V .

- (i) A *Dirichlet condition* specifies the value of φ on ∂V .
- (ii) A *Neumann condition* specifies the normal derivative $\mathbf{n} \cdot \nabla\varphi$ (also written $\frac{\partial\varphi}{\partial\mathbf{n}}$) on ∂V .

Remark. The appropriate boundary condition depends on the physical problem. For example, specifying $\frac{\partial\varphi}{\partial\mathbf{n}}$ corresponds to prescribing the normal component of $\mathbf{g}$ or $\mathbf{E}$. Different boundary conditions may be imposed on different components of ∂V .

Example (Spherical shell Laplace equation). We might have a spherically symmetric distribution with constant ρ_0 , defined in $a \leq r \leq b$, with $\varphi(a) = 0$ and $\frac{\partial\varphi}{\partial\mathbf{n}}(b) = 0$.

Then the general solution is

$$\varphi(r) = \frac{A}{r} + B - \frac{1}{6}\rho_0 r^2.$$

We apply the first boundary condition to obtain

$$\frac{A}{a} + B - \frac{1}{6}\rho_0 a^2 = 0.$$

The second boundary condition gives

$$\mathbf{n} \cdot \nabla\varphi = -\frac{A}{b^2} - \frac{1}{3}\rho_0 b = 0.$$

These conditions give

$$A = -\frac{1}{3}\rho_0 b^3, \quad B = \frac{1}{6}\rho_0 a^2 + \frac{1}{3}\rho_0 \frac{b^3}{a}.$$

Example (Spherically symmetric Poisson solution). We might also be interested with spherically symmetric solution with

$$\nabla^2 \varphi = \begin{cases} -\rho_0 & r \leq a \\ 0 & r > a \end{cases}$$

with φ non-singular at $r = 0$ and $\varphi(r) \rightarrow 0$ as $r \rightarrow \infty$, and φ, φ' continuous at $r = a$. This models the gravitational potential on a uniform planet.

Then the general solution from above is

$$\varphi = \begin{cases} \frac{A}{r} + B - \frac{1}{6}\rho_0 r^2 & r \leq a \\ \frac{C}{r} + D & r > a. \end{cases}$$

Since φ is non-singular at $r = 0$, we have $A = 0$. Since $\varphi \rightarrow 0$ as $r \rightarrow \infty$, $D = 0$. So

$$\varphi = \begin{cases} B - \frac{1}{6}\rho_0 r^2 & r \leq a \\ \frac{C}{r} & r > a. \end{cases}$$

This is the gravitational potential inside and outside a planet of constant density ρ_0 and radius a . We want φ and φ' to be continuous at $r = a$. So we have

$$\begin{aligned} B + \frac{1}{6}4\pi\rho_0 G a^2 &= \frac{C}{a} \\ \frac{4}{3}\pi G \rho_0 a &= -\frac{C}{a^2}. \end{aligned}$$

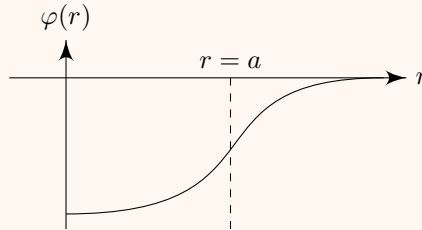
The second equation gives $C = -GM$. Substituting that into the first equation to find B , we get

$$\varphi(r) = \begin{cases} \frac{GM}{2a} \left[\left(\frac{r}{a} \right)^2 - 3 \right] & r \leq a \\ -\frac{GM}{r} & r > a \end{cases}$$

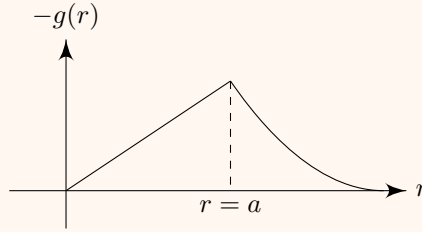
Since $g = -\varphi'$, we have

$$g(r) = \begin{cases} -\frac{GM r}{a^3} & r \leq a \\ -\frac{GM}{r^2} & r > a \end{cases}$$

We can plot the potential energy:



We can also plot $-g(r)$, the inward acceleration:



Alternatively, we can apply Gauss' Law for a flux of $\mathbf{g} = g(r)\mathbf{e}_r$ out of S , a sphere of radius R . For $R \leq a$,

$$\int_S \mathbf{g} \cdot d\mathbf{S} = 4\pi R^2 g(R) = -4\pi GM \left(\frac{R}{a}\right)^3$$

So

$$g(R) = -\frac{GMR}{a^3}.$$

For $R \geq a$, we can simply apply Newton's law of gravitation.

Remark (Gauss flux method). For any Poisson equation $\nabla^2 \varphi = -\rho$ with sufficient symmetry (not necessarily gravitational or electrostatic), one can exploit the divergence theorem:

$$\int_S \nabla \varphi \cdot d\mathbf{S} = - \int_V \rho \, dV.$$

Choosing S to respect the symmetry of the problem often simplifies the computation. This approach is known as the *Gauss flux method*.

11 Laplace's and Poisson's equations

11.1 Uniqueness theorems

Theorem (Uniqueness theorem for Poisson's equation). Consider $\nabla^2 \varphi = -\rho$ for some $\rho(\mathbf{r})$ on a bounded volume V with $S = \partial V$ being a closed surface, with an outward normal $\mathbf{n}$.

Suppose φ satisfies either

- (i) Dirichlet condition, $\varphi(\mathbf{r}) = f(\mathbf{r})$ on S
- (ii) Neumann condition $\frac{\partial \varphi(\mathbf{r})}{\partial \mathbf{n}} = \mathbf{n} \cdot \nabla \varphi = g(\mathbf{r})$ on S .

where f, g are given. Then

- (i) $\varphi(\mathbf{r})$ is unique
- (ii) $\varphi(\mathbf{r})$ is unique up to a constant.

Remark. This theorem has significant practical value: if a solution is found by any method (symmetry arguments, guessing, numerical computation), uniqueness guarantees it is the only solution (up to a constant in the Neumann case).

Proof. The proofs for both boundary conditions follow the same structure. Let $\varphi_1(\mathbf{r})$ and $\varphi_2(\mathbf{r})$ satisfy Poisson's equation, each obeying the boundary conditions (N) or (D). Then $\Psi(\mathbf{r}) = \varphi_2(\mathbf{r}) - \varphi_1(\mathbf{r})$ satisfies $\nabla^2 \Psi = 0$ on V by linearity, and

- (i) $\Psi = 0$ on S ; or
- (ii) $\frac{\partial \Psi}{\partial \mathbf{n}} = 0$ on S .

Combining these two together, we know that $\Psi \frac{\partial \Psi}{\partial \mathbf{n}} = 0$ on the surface. So using the divergence theorem,

$$\int_V \nabla \cdot (\Psi \nabla \Psi) \, dV = \int_S (\Psi \nabla \Psi) \cdot d\mathbf{S} = 0.$$

But

$$\nabla \cdot (\Psi \nabla \Psi) = (\nabla \Psi) \cdot (\nabla \Psi) + \underbrace{\Psi \nabla^2 \Psi}_{=0} = |\nabla \Psi|^2.$$

So

$$\int_V |\nabla \Psi|^2 \, dV = 0.$$

Since $|\nabla \Psi|^2 \geq 0$, the integral can only vanish if $|\nabla \Psi| = 0$. So $\nabla \Psi = 0$. So $\Psi = c$, a constant on V . So

- (i) $\Psi = 0$ on $S \Rightarrow c = 0$. So $\varphi_1 = \varphi_2$ on V .
- (ii) $\varphi_2(\mathbf{r}) = \varphi_1(\mathbf{r}) + C$, as claimed. □

Remark (Existence and compatibility conditions). Uniqueness does not guarantee existence. For Neumann boundary conditions, integrating Poisson's equation over V and applying the divergence theorem yields the *compatibility condition*

$$\int_V \rho \, dV + \int_{\partial V} g \, dS = 0.$$

If ρ and g do not satisfy this condition, no solution exists.

Remark (Extensions). The uniqueness theorem generalizes to regions in $\mathbb{R}^n$ and to unbounded domains. For unbounded domains, one imposes decay conditions such as $|\Psi(\mathbf{r})| = O(1/R)$ or $|\frac{\partial \Psi}{\partial n}(\mathbf{r})| = O(1/R^2)$ as $R \rightarrow \infty$. Similar results hold for mixed boundary conditions (Dirichlet on part of ∂V , Neumann on the rest), though each case requires separate verification.

The proof uses a special case of Green's first identity.

Proposition (Green's first identity). For scalar fields u and v on a region V with boundary S ,

$$\int_S (u \nabla v) \cdot d\mathbf{S} = \int_V (\nabla u) \cdot (\nabla v) \, dV + \int_V u \nabla^2 v \, dV.$$

Proposition (Green's second identity). By interchanging u and v in Green's first identity and subtracting,

$$\int_S (u \nabla v - v \nabla u) \cdot d\mathbf{S} = \int_V (u \nabla^2 v - v \nabla^2 u) \, dV.$$

Remark. Both identities follow directly from the divergence theorem applied to $u \nabla v$ and $v \nabla u$.

11.2 Laplace's equation and harmonic functions

Definition (Harmonic function). A *harmonic function* is a solution to Laplace's equation $\nabla^2 \varphi = 0$.

Harmonic functions possess several remarkable properties.

11.2.1 The mean value property

Proposition (Mean value property). Suppose $\varphi(\mathbf{r})$ is harmonic on region V containing a solid sphere defined by $|\mathbf{r} - \mathbf{a}| \leq R$, with boundary $S_R = |\mathbf{r} - \mathbf{a}| = R$, for some R . Define

$$\bar{\varphi}(R) = \frac{1}{4\pi R^2} \int_{S_R} \varphi(\mathbf{r}) \, dS.$$

Then $\varphi(\mathbf{a}) = \bar{\varphi}(R)$. That is, the value of a harmonic function at any point equals its average over any sphere centered at that point.

Proof. Note that $\bar{\varphi}(R) \rightarrow \varphi(\mathbf{a})$ as $R \rightarrow 0$. We take spherical coordinates (u, θ, χ) centered on $\mathbf{r} = \mathbf{a}$. The scalar element (when $u = R$) on S_R is

$$dS = R^2 \sin \theta \, d\theta \, d\chi.$$

So $\frac{dS}{R^2}$ is independent of R . Write

$$\bar{\varphi}(R) = \frac{1}{4\pi} \int \varphi \frac{dS}{R^2}.$$

Differentiate this with respect to R , noting that dS/R^2 is independent of R . Then we obtain

$$\frac{d}{dR} \bar{\varphi}(R) = \frac{1}{4\pi R^2} \int \left. \frac{\partial \varphi}{\partial u} \right|_{u=R} dS$$

But

$$\frac{\partial \varphi}{\partial u} = \mathbf{e}_u \cdot \nabla \varphi = \mathbf{n} \cdot \nabla \varphi = \frac{\partial \varphi}{\partial \mathbf{n}}$$

on S_R . So

$$\frac{d}{dR} \bar{\varphi}(R) = \frac{1}{4\pi R^2} \int_{S_R} \nabla \varphi \cdot d\mathbf{S} = \frac{1}{4\pi R^2} \int_{V_R} \nabla^2 \varphi \, dV = 0$$

by divergence theorem. So $\bar{\varphi}(R)$ does not depend on R , and the result follows. $\square$

11.2.2 The maximum (or minimum) principle

The following results are stated for maxima; analogous statements hold for minima.

Definition (Local maximum). We say that $\varphi(\mathbf{r})$ has a *local maximum* at $\mathbf{a}$ if for some $\varepsilon > 0$, $\varphi(\mathbf{r}) < \varphi(\mathbf{a})$ when $0 < |\mathbf{r} - \mathbf{a}| < \varepsilon$.

Proposition (Maximum principle). If a function φ is harmonic on a region V , then φ cannot have a maximum at an interior point $\mathbf{a}$ of V .

Proof. Suppose that φ had a local maximum at $\mathbf{a}$ in the interior. Then there is an ε such that for any $\mathbf{r}$ such that $0 < |\mathbf{r} - \mathbf{a}| < \varepsilon$, we have $\varphi(\mathbf{r}) < \varphi(\mathbf{a})$.

Note that if there is an ε that works, then any smaller ε will work. Pick an ε sufficiently small such that the region $|\mathbf{r} - \mathbf{a}| < \varepsilon$ lies within V (possible since $\mathbf{a}$ lies in the interior of V).

Then for any $\mathbf{r}$ such that $|\mathbf{r} - \mathbf{a}| = \varepsilon$, we have $\varphi(\mathbf{r}) < \varphi(\mathbf{a})$.

$$\bar{\varphi}(\varepsilon) = \frac{1}{4\pi\varepsilon^2} \int_{S_\varepsilon} \varphi(\mathbf{r}) \, dS < \varphi(\mathbf{a}),$$

which contradicts the mean value property. $\square$

Remark (Local analysis of stationary points). The maximum principle can also be understood via the Hessian. At a stationary point $\mathbf{a}$ where $\nabla\varphi = \mathbf{0}$, let λ_i denote the eigenvalues of the Hessian matrix $H_{ij} = \frac{\partial^2\varphi}{\partial x_i \partial x_j}$. Since φ is harmonic, $\nabla^2\varphi = \text{tr}(H) = \sum_i \lambda_i = 0$. A local maximum or minimum requires all eigenvalues to have the same sign, which is impossible if their sum vanishes. Thus stationary points of harmonic functions can only be saddle points (assuming at least one $\lambda_i \neq 0$).

11.3 Integral solutions of Poisson's equation

11.3.1 Statement and heuristic derivation

We seek explicit solutions to Poisson's equation by generalizing from discrete to continuous sources.

For a single point source of strength λ at $\mathbf{a}$, the potential is

$$\varphi = \frac{\lambda}{4\pi} \frac{1}{|\mathbf{r} - \mathbf{a}|}.$$

Remark. For gravitation, $\lambda = -4\pi GM$; for electrostatics, $\lambda = Q/\varepsilon_0$.

For multiple point sources λ_α at positions $\mathbf{r}_\alpha$, linearity gives

$$\varphi(\mathbf{r}) = \sum_\alpha \frac{1}{4\pi} \frac{\lambda_\alpha}{|\mathbf{r} - \mathbf{r}_\alpha|}.$$

For a continuous source distribution $\rho(\mathbf{r})$, replacing the sum by an integral suggests:

Proposition (Poisson's equation integral solution). The solution to Poisson's equation $\nabla^2\varphi = -\rho$, with boundary conditions $|\varphi(\mathbf{r})| = O(1/|\mathbf{r}|)$ and $|\nabla\varphi(\mathbf{r})| = O(1/|\mathbf{r}|^2)$, is

$$\varphi(\mathbf{r}) = \frac{1}{4\pi} \int_{V'} \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, dV'$$

For $\rho(\mathbf{r}')$ non-zero everywhere, but suitably well-behaved as $|\mathbf{r}'| \rightarrow \infty$, we can also take $V' = \mathbb{R}^3$.

Example (Poisson integral, uniform ball). Suppose

$$\nabla^2 \varphi = \begin{cases} -\rho_0 & |\mathbf{r}| \leq a \\ 0 & |\mathbf{r}| > a. \end{cases}$$

Fix $\mathbf{r}$ and introduce polar coordinates r', θ, χ for $\mathbf{r}'$. We take the $\theta = 0$ direction to be the direction along the line from $\mathbf{r}'$ to $\mathbf{r}$.

Then

$$\varphi(\mathbf{r}) = \frac{1}{4\pi} \int_{V'} \frac{\rho_0}{|\mathbf{r} - \mathbf{r}'|} dV'.$$

We have

$$dV' = r'^2 \sin \theta \, dr' \, d\theta \, d\chi.$$

We also have

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{r^2 + r'^2 - 2rr' \cos \theta}$$

by the cosine rule ($c^2 = a^2 + b^2 - 2ab \cos C$). So

$$\begin{aligned} \varphi(\mathbf{r}) &= \frac{1}{4\pi} \int_0^a dr' \int_0^\pi d\theta \int_0^{2\pi} d\chi \frac{\rho_0 r'^2 \sin \theta}{\sqrt{r^2 + r'^2 - 2rr' \cos \theta}} \\ &= \frac{\rho_0}{2} \int_0^a dr' \frac{r'^2}{r'} \left[\sqrt{r^2 + r'^2 - 2rr' \cos \theta} \right]_{\theta=0}^{\theta=\pi} \\ &= \frac{\rho_0}{2} \int_0^a dr' \frac{r'}{r} (|\mathbf{r} + \mathbf{r}'| - |\mathbf{r} - \mathbf{r}'|) \\ &= \frac{\rho_0}{2} \int_0^a \left[dr' \frac{r'}{r} \left(\begin{cases} 2r' & r > r' \\ 2r & r < r' \end{cases} \right) \right] \end{aligned}$$

If $r > a$, then $r > r'$ always. So

$$\varphi(\mathbf{r}) = \rho_0 \int_0^a \frac{r'^2}{r} dr' = \frac{\rho_0 a^3}{3r}.$$

If $r < a$, then the integral splits into two parts:

$$\varphi(\mathbf{r}) = \rho_0 \left(\int_0^r dr' \frac{r'^2}{r} + \int_r^a dr' r' \right) = \rho_0 \left[-\frac{1}{6} r^2 + \frac{a^2}{2} \right].$$

11.3.2 Point sources and δ -functions*

Consider the point source potential

$$\Psi = \frac{\lambda}{4\pi |\mathbf{r} - \mathbf{a}|}.$$

For $\mathbf{r} \neq \mathbf{a}$, direct computation gives

$$\nabla \Psi = -\frac{\lambda}{4\pi} \frac{\mathbf{r} - \mathbf{a}}{|\mathbf{r} - \mathbf{a}|^3}, \quad \nabla^2 \Psi = 0.$$

At $\mathbf{r} = \mathbf{a}$, the potential Ψ is singular. To characterize $\nabla^2 \Psi$ at this point, we use the divergence theorem.

For any sphere with center $\mathbf{a}$, we have

$$\int_S \nabla \Psi \cdot d\mathbf{S} = -\lambda.$$

By the divergence theorem, we have

$$\int \nabla^2 \Psi \, dV = -\lambda.$$

for V being a solid sphere with $\partial V = S$. Since $\nabla^2 \Psi$ is zero at any point $\mathbf{r} \neq \mathbf{a}$, we must have

$$\nabla^2 \Psi = -\lambda \delta(\mathbf{r} - \mathbf{a}),$$

where δ is the 3d delta function, which satisfies

$$\int_V f(\mathbf{r}) \delta(\mathbf{r} - \mathbf{a}) \, dV = f(\mathbf{a})$$

for any volume containing $\mathbf{a}$.

In short, we have

$$\nabla^2 \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) = -4\pi \delta(\mathbf{r} - \mathbf{r}').$$

Using these, we can verify that the integral solution of Poisson's equation we obtained previously is correct:

$$\begin{aligned} \nabla^2 \Psi(\mathbf{r}) &= \nabla^2 \left(\frac{1}{4\pi} \int_{V'} \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, dV' \right) \\ &= \frac{1}{4\pi} \int_{V'} \rho(\mathbf{r}') \nabla^2 \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) \, dV' \\ &= - \int_{V'} \rho(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') \, dV' \\ &= -\rho(\mathbf{r}), \end{aligned}$$

as required.

12 Maxwell's equations

12.1 Laws of electromagnetism

Maxwell's equations are four partial differential equations that govern classical electromagnetism. Together with the Lorentz force law, they form a complete description of electromagnetic phenomena; all other results in classical electromagnetism are mathematical consequences of these equations.

Electromagnetism is described by two vector fields: the *electric field* $\mathbf{E}(\mathbf{r}, t)$ and the *magnetic field* $\mathbf{B}(\mathbf{r}, t)$. The effect of these fields on charged particles is given by the Lorentz force law.

Law (Lorentz force law). A point charge q experiences a force of

$$\mathbf{F} = q(\mathbf{E} + \dot{\mathbf{r}} \times \mathbf{B}).$$

The dynamics of the field itself is governed by *Maxwell's equations*. To state the equations, we need to introduce two more concepts.

Definition (Charge and current density). $\rho(\mathbf{r}, t)$ is the *charge density*, defined as the charge per unit volume.

$\mathbf{j}(\mathbf{r}, t)$ is the *current density*, defined as the electric current per unit area of cross section.

Then Maxwell's equations say

Law (Maxwell's equations).

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0 \\ \nabla \times \mathbf{B} - \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \mathbf{j},\end{aligned}$$

where ε_0 is the electric constant (permittivity of free space) and μ_0 is the magnetic constant (permeability of free space), which are constants determined experimentally.

Several fundamental results follow directly from Maxwell's equations.

Proposition (Conservation of charge). Taking the divergence of the fourth Maxwell equation yields the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0.$$

Proof. We have

$$\underbrace{\nabla \cdot (\nabla \times \mathbf{B})}_{=0} - \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \underbrace{(\nabla \cdot \mathbf{E})}_{=\rho/\varepsilon_0} = \mu_0 \nabla \cdot \mathbf{j}.$$

Rearranging gives the result. $\square$

Proposition (Integral forms of Maxwell's equations). Integrating the differential forms and applying the divergence and Stokes' theorems yields:

$$(i) \int_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\varepsilon_0} \quad (\text{Gauss' law for electric fields})$$

$$(ii) \int_S \mathbf{B} \cdot d\mathbf{S} = 0 \quad (\text{no magnetic monopoles})$$

$$(iii) \oint_C \mathbf{E} \cdot d\mathbf{r} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S} \quad (\text{Faraday's law})$$

Remark. Faraday's law states that a time-varying magnetic flux through a surface induces an electromotive force around its boundary.

12.2 Static charges and steady currents

When ρ , $\mathbf{j}$, $\mathbf{E}$, and $\mathbf{B}$ are all independent of time, Maxwell's equations decouple into separate equations for the electric and magnetic fields.

For the electric field, we have

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \rho/\varepsilon_0 \\ \nabla \times \mathbf{E} &= \mathbf{0}. \end{aligned}$$

The second equation implies $\mathbf{E} = -\nabla\varphi$ for some scalar potential φ . Substituting into the first gives Poisson's equation: $\nabla^2\varphi = -\rho/\varepsilon_0$.

For the magnetic field, we have

$$\begin{aligned} \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{j}. \end{aligned}$$

The first equation implies $\mathbf{B} = \nabla \times \mathbf{A}$ for some *vector potential* $\mathbf{A}$.

Remark (Gauge freedom). The vector potential is not unique: the transformation $\mathbf{A} \mapsto \mathbf{A} + \nabla\chi$ for any scalar field χ leaves $\mathbf{B}$ unchanged, since $\nabla \times (\nabla\chi) = \mathbf{0}$. Choosing χ such that $\nabla \cdot \mathbf{A} = 0$ (the *Coulomb gauge*) yields

$$\nabla^2 \mathbf{A} = \nabla(\underbrace{\nabla \cdot \mathbf{A}}_{=0}) - \nabla \times (\underbrace{\nabla \times \mathbf{A}}_{=\mathbf{B}}) = -\mu_0 \mathbf{j}.$$

The following table summarizes electrostatics and magnetostatics:

Electrostatics	Magnetostatics
$\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$	$\nabla \cdot \mathbf{B} = 0$
$\nabla \times \mathbf{E} = \mathbf{0}$	$\nabla \times \mathbf{B} = \mu_0 \mathbf{j}$
$\nabla^2 \varphi = -\rho/\varepsilon_0$	$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{j}$
ε_0 sets the scale of electrostatic effects, e.g. the Coulomb force	μ_0 sets the scale of magnetic effects, e.g. force between two wires with cur- rents.

12.3 Electromagnetic waves

Consider Maxwell's equations in free space, where $\rho = 0$ and $\mathbf{j} = \mathbf{0}$.

Proposition (Wave equations for electromagnetic fields). In free space, both $\mathbf{E}$ and $\mathbf{B}$ satisfy the wave equation

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{E} = \mathbf{0}, \quad \left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{B} = \mathbf{0},$$

where $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

Proof. Using the vector identity $\nabla^2 \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla \times (\nabla \times \mathbf{E})$ and Maxwell's equations:

$$\nabla^2 \mathbf{E} = \nabla(\underbrace{\nabla \cdot \mathbf{E}}_{=0}) - \nabla \times (\nabla \times \mathbf{E}) = \nabla \times \frac{\partial \mathbf{B}}{\partial t} = \frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

The equation for $\mathbf{B}$ follows similarly. □

Remark. Maxwell's equations thus predict the existence of electromagnetic waves propagating at speed $c = 1/\sqrt{\varepsilon_0 \mu_0} \approx 3.00 \times 10^8 \text{ m s}^{-1}$. This value coincides with the measured speed of light, leading Maxwell to conclude that light is an electromagnetic wave.

13 Tensors and tensor fields

13.1 Definition

A vector in $\mathbb{R}^3$ can be viewed geometrically as a point in space, or algebraically as a list of three numbers. The latter is a *representation* with respect to a chosen basis. Under a change of basis described by a rotation matrix R , the components transform as

$$v'_i = R_{ip}v_p.$$

Similarly, a matrix representing a linear transformation has components that transform as

$$A'_{ij} = R_{ip}R_{jq}A_{pq}.$$

For an array of numbers A_{ij} to represent a geometrically meaningful object (such as a linear transformation), its components in different bases must be related by the appropriate transformation law. An arbitrary assignment of numbers in each basis—say $A_{ij} = \delta_{ij}$ in one basis and $A_{ij} = 0$ in another—would not correspond to any geometric object.

Objects whose components transform according to specific rules under change of basis are called *tensors*. The transformation rule generalizes naturally to objects with any number of indices:

$$T'_{ij\dots k} = R_{ip}R_{jq}\cdots R_{kr}T_{pq\dots r}.$$

Definition (Tensor). A *tensor* of rank n has components $T_{ij\dots k}$ (with n indices) with respect to each basis $\{\mathbf{e}_i\}$ or coordinate system $\{x_i\}$, and satisfies the following rule of change of basis:

$$T'_{ij\dots k} = R_{ip}R_{jq}\cdots R_{kr}T_{pq\dots r}.$$

Example (Tensors of rank 0, 1, 2).

- A tensor of rank 0 has no indices and is invariant under change of basis: a *scalar*.
- A tensor of rank 1 transforms as $T'_i = R_{ip}T_p$: a *vector*.
- A tensor of rank 2 transforms as $T'_{ij} = R_{ip}R_{jq}T_{pq}$: a *matrix*.

Example (Constructing tensors from vectors).

- (i) If $\mathbf{u}, \mathbf{v}, \dots, \mathbf{w}$ are n vectors, then $T_{ij\dots k} = u_iv_j\cdots w_k$ defines a tensor of rank n . For example, with $n = 2$:

$$T'_{ij} = u'_iv'_j = (R_{ip}u_p)(R_{jq}v_q) = R_{ip}R_{jq}T_{pq}.$$

Linear combinations of such products are also tensors, e.g., $T_{ij} = u_iv_j + a_ib_j$.

- (ii) The Kronecker delta δ_{ij} and Levi-Civita symbol ε_{ijk} are tensors of

rank 2 and 3 whose components are the same in every basis:

$$\delta'_{ij} = R_{ip}R_{jq}\delta_{pq} = R_{ip}R_{jp} = (RR^T)_{ij} = \delta_{ij},$$

and

$$\varepsilon'_{ijk} = R_{ip}R_{jq}R_{kr}\varepsilon_{pqr} = (\det R)\varepsilon_{ijk} = \varepsilon_{ijk}.$$

- (iii) In anisotropic materials, an applied electric field $\mathbf{E}$ produces a current density $\mathbf{j}$ via the linear relation $j_i = \sigma_{ij}E_j$, where σ_{ij} is the *conductivity tensor*. The current need not be parallel to the field due to preferential crystallographic directions. For *isotropic* materials, $\sigma_{ij} = \sigma\delta_{ij}$ and the current is parallel to the field.

13.2 Tensor algebra

Definition (Tensor addition). Tensors T and S of the same rank can be *added*: $(T + S)_{ij\dots k} = T_{ij\dots k} + S_{ij\dots k}$.

The sum is again a tensor, since $(T + S)'_{ij} = R_{ip}R_{jq}(T_{pq} + S_{pq})$.

Definition (Scalar multiplication). A tensor T of rank n multiplied by a scalar α gives a tensor of the same rank: $(\alpha T)_{ij\dots k} = \alpha T_{ij\dots k}$.

Definition (Tensor product). Let T be a tensor of rank n and S be a tensor of rank m . The *tensor product* $T \otimes S$ is a tensor of rank $n + m$ defined by

$$(T \otimes S)_{x_1x_2\dots x_ny_1y_2\dots y_m} = T_{x_1x_2\dots x_n}S_{y_1y_2\dots y_m}.$$

Remark. In particular, for vectors $\mathbf{u}, \mathbf{v}, \dots, \mathbf{w}$, the tensor product $\mathbf{u} \otimes \mathbf{v} \otimes \dots \otimes \mathbf{w}$ has components $u_iv_j \dots w_k$.

Definition (Tensor contraction). For a tensor T of rank n with components $T_{ijp\dots q}$, *contraction* on indices i and j produces a tensor of rank $n - 2$:

$$S_{p\dots q} = \delta_{ij}T_{ijp\dots q} = T_{iip\dots q}.$$

Contraction can be performed on any pair of indices.

Remark. Contraction preserves tensor character. For a rank-2 tensor T_{ij} , contracting gives the scalar $T'_{ii} = R_{ip}R_{iq}T_{pq} = \delta_{pq}T_{pq} = T_{ii}$, using orthogonality of R . Viewing T_{ij} as a matrix, contraction is the trace, confirming that the trace is basis-independent.

Remark. Matrix multiplication can be expressed as a tensor product followed by contraction: $(MN)_{iq} = M_{ij}N_{jq}$.

13.3 Symmetric and antisymmetric tensors

Definition (Symmetric and anti-symmetric tensors). A tensor T of rank n is *symmetric* in the indices i, j if it obeys

$$T_{ijp\dots q} = T_{jip\dots q}.$$

It is *anti-symmetric* if

$$T_{ijp\dots q} = -T_{jip\dots q}.$$

Again, a tensor can be symmetric or anti-symmetric in any pair of indices, not just the first two.

Remark. Symmetry properties are preserved under change of basis:

$$T'_{klr\dots s} = R_{ki}R_{lj}R_{rp}\cdots R_{sq}T_{ijp\dots q} = \pm R_{ki}R_{lj}R_{rp}\cdots R_{sq}T_{jip\dots q} = \pm T'_{lkr\dots s}.$$

Definition (Totally symmetric and anti-symmetric tensors). A tensor is *totally (anti-)symmetric* if it is (anti-)symmetric in every pair of indices.

Example (Symmetric and antisymmetric tensors). $\delta_{ij} = \delta_{ji}$ is totally symmetric, while $\varepsilon_{ijk} = -\varepsilon_{jik}$ is totally antisymmetric.

Proposition (Totally antisymmetric tensors in $\mathbb{R}^3$).

- (i) Any totally antisymmetric rank-3 tensor is a scalar multiple of ε_{ijk} .
- (ii) There are no non-zero totally antisymmetric tensors of rank greater than 3.

Proof. For (ii): a totally antisymmetric tensor with more than 3 indices in $\mathbb{R}^3$ must have a repeated index value (by the pigeonhole principle), forcing the component to vanish. $\square$

13.4 Tensors, multi-linear maps and the quotient rule

Tensors as multi-linear maps

Definition (Multilinear map). A map $T : \underbrace{\mathbb{R}^3 \times \cdots \times \mathbb{R}^3}_n \rightarrow \mathbb{R}$ is *multi-linear* if it is linear in each argument separately, i.e. for each $k = 1, \dots, n$ and all scalars λ, μ ,

$$T(\dots, \lambda \mathbf{u} + \mu \mathbf{v}, \dots) = \lambda T(\dots, \mathbf{u}, \dots) + \mu T(\dots, \mathbf{v}, \dots),$$

where linearity is in the k th argument with all other arguments held fixed.

Theorem (Tensors are multilinear maps). There is a one-to-one correspondence between rank- n tensors and multi-linear maps from n vectors to $\mathbb{R}$. Explicitly, a rank- n tensor $T_{ij\dots k}$ defines a multi-linear map by

$$T(\mathbf{a}, \mathbf{b}, \dots, \mathbf{c}) = T_{ij\dots k} a_i b_j \cdots c_k,$$

and conversely every multi-linear map arises in this way.

Proof. We must show two things.

(i) *A tensor defines a well-defined multi-linear map.* The expression $T_{ij\dots k} a_i b_j \cdots c_k$ is obtained from the tensor product of $T_{ij\dots k}$ with $a_i, b_j, \dots, c_k$, followed by contraction. Since tensor products and contractions preserve tensor-ness, the result is a rank-0 tensor, i.e. a scalar, which is independent of the basis chosen. Multi-linearity is immediate from the summation convention.

(ii) *A multi-linear map determines a tensor.* Given a multi-linear map T , define $T_{ij\dots k} = T(\mathbf{e}_i, \mathbf{e}_j, \dots, \mathbf{e}_k)$ in any basis $\{\mathbf{e}_i\}$. We must show that $T_{ij\dots k}$ obeys the tensor transformation rule. Since T is a map independent of the basis, we have

$$T_{ij\dots k} a_i b_j \cdots c_k = T'_{ij\dots k} a'_i b'_j \cdots c'_k.$$

Since $a_i = R_{pi} a'_p$ (inverting the transformation rule $a'_p = R_{pi} a_i$), substituting gives

$$T_{ij\dots k} (R_{pi} a'_p) (R_{qj} b'_q) \cdots (R_{rk} c'_r) = T'_{pq\dots r} a'_p b'_q \cdots c'_r.$$

Since this holds for all vectors $\mathbf{a}, \mathbf{b}, \dots, \mathbf{c}$, we conclude

$$T'_{pq\dots r} = R_{pi} R_{qj} \cdots R_{rk} T_{ij\dots k},$$

which is the tensor transformation rule. $\square$

Remark. This viewpoint defines tensors independently of any coordinate system; the transformation rule then emerges as a consequence. The same approach relates matrices to linear maps.

The quotient rule

Proposition (Contraction of tensors). If $T_{i\dots j p\dots q}$ is a tensor of rank $n + m$ and $u_{p\dots q}$ is a tensor of rank m , then

$$v_{i\dots j} = T_{i\dots j p\dots q} u_{p\dots q}$$

is a tensor of rank n , since it is a tensor product of T and u followed by contraction.

Proposition (Quotient rule). Suppose that $T_{i\dots j p\dots q}$ is an array of numbers defined in each coordinate system, and that $v_{i\dots j} = T_{i\dots j p\dots q} u_{p\dots q}$ is a tensor of rank n for every tensor $u_{p\dots q}$ of rank m . Then $T_{i\dots j p\dots q}$ is a tensor of rank $n + m$.

Remark. The special case $n = m = 1$ recovers the earlier result that if $v_i = T_{ij} u_j$ is a vector for every vector u_j , then T_{ij} is a rank-2 tensor. That is, linear maps are tensors.

Proof of the quotient rule. Consider the special form $u_{p\dots q} = c_p \cdots d_q$ for arbitrary vectors $\mathbf{c}, \dots, \mathbf{d}$. By assumption,

$$v_{i\dots j} = T_{i\dots j p\dots q} c_p \cdots d_q$$

is a tensor. Then for any vectors $\mathbf{a}, \dots, \mathbf{b}$,

$$v_{i\dots j} a_i \cdots b_j = T_{i\dots j p\dots q} a_i \cdots b_j c_p \cdots d_q$$

is a scalar for all choices of $\mathbf{a}, \dots, \mathbf{b}, \mathbf{c}, \dots, \mathbf{d}$. Since a scalar is basis-independent, $T_{i\dots j p\dots q}$ defines a multi-linear map. By the preceding theorem, $T_{i\dots j p\dots q}$ is a tensor. $\square$

13.5 Tensor calculus**Tensor fields and derivatives**

Definition (Tensor field). A *tensor field* assigns a tensor $T_{ij\dots k}(\mathbf{x})$ to each point $\mathbf{x}$ in space.

Proposition (Tensor field derivative rank). Let $T_{ij\dots k}(\mathbf{x})$ be a smooth tensor field of rank n . Then

$$\underbrace{\frac{\partial}{\partial x_p} \cdots \frac{\partial}{\partial x_q}}_m T_{ij\dots k} \quad (*)$$

is a tensor of rank $n + m$.

Proof. It suffices to show that $\frac{\partial}{\partial x_p}$ transforms as a rank-1 tensor. Since $x'_i = R_{iq}x_q$, we have $\frac{\partial x_q}{\partial x'_i} = R_{iq}$. By the chain rule,

$$\frac{\partial}{\partial x'_i} = \frac{\partial x_q}{\partial x'_i} \frac{\partial}{\partial x_q} = R_{iq} \frac{\partial}{\partial x_q}.$$

Thus $\frac{\partial}{\partial x_p}$ obeys the vector transformation rule, and the result follows by the same argument as for tensor products. $\square$

Integrals and the tensor divergence theorem

Definition (Tensor-valued integral). Tensor-valued integrals are defined componentwise. For a tensor field $T_{ij\dots k}(\mathbf{x})$ of rank n , the integral $\int_V T_{ij\dots k} dV$ is a tensor of the same rank.

Example (Momentum flux tensor). For a fluid with velocity field $\mathbf{u}(\mathbf{x})$ and density ρ , the flux of the i th component of momentum through a surface element is $\rho u_i u_j n_j \delta S = T_{ij} n_j \delta S$, where $T_{ij} = \rho u_i u_j$ is the momentum flux tensor.

Theorem (Divergence theorem for tensors). Let V be a volume in $\mathbb{R}^3$ bounded by a surface $S = \partial V$ with outward unit normal $\mathbf{n}$, and let $T_{ij\dots k\ell}(\mathbf{x})$ be a smooth tensor field. Then

$$\int_S T_{ij\dots k\ell} n_\ell dS = \int_V \frac{\partial}{\partial x_\ell} (T_{ij\dots k\ell}) dV.$$

Remark. The standard divergence theorem is the special case where T is a vector field (rank 1).

Proof. Apply the usual divergence theorem to the vector field $\mathbf{v}$ defined by $v_\ell = a_i b_j \cdots c_k T_{ij\dots k\ell}$, where $\mathbf{a}, \mathbf{b}, \dots, \mathbf{c}$ are fixed constant vectors. Then

$$\nabla \cdot \mathbf{v} = \frac{\partial v_\ell}{\partial x_\ell} = a_i b_j \cdots c_k \frac{\partial}{\partial x_\ell} T_{ij\dots k\ell},$$

and

$$\mathbf{n} \cdot \mathbf{v} = n_\ell v_\ell = a_i b_j \cdots c_k T_{ij\dots k\ell} n_\ell.$$

Since $\mathbf{a}, \mathbf{b}, \dots, \mathbf{c}$ are arbitrary, they can be eliminated, and the tensor divergence theorem follows. $\square$

14 Tensors of rank 2

14.1 Decomposition of a second-rank tensor

Proposition (Symmetric-antisymmetric decomposition). Any rank-2 tensor T_{ij} in $\mathbb{R}^3$ can be uniquely decomposed as

$$T_{ij} = S_{ij} + A_{ij},$$

where $S_{ij} = \frac{1}{2}(T_{ij} + T_{ji})$ is symmetric and $A_{ij} = \frac{1}{2}(T_{ij} - T_{ji})$ is antisymmetric.

Remark (Component counting). A general rank-2 tensor T_{ij} in $\mathbb{R}^3$ has 9 independent components. The symmetric part S_{ij} has 6 and the antisymmetric part A_{ij} has 3, since they must be of the form

$$(S_{ij}) = \begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}, \quad (A_{ij}) = \begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}.$$

Proposition (Traceless-isotropic decomposition). Any symmetric rank-2 tensor S_{ij} can be further decomposed as

$$S_{ij} = P_{ij} + \frac{1}{3}\delta_{ij}Q,$$

where $Q = S_{ii}$ is the trace and $P_{ij} = S_{ij} - \frac{1}{3}\delta_{ij}Q$ is a symmetric traceless tensor. The isotropic part $\frac{1}{3}\delta_{ij}Q$ has 1 independent component, while the traceless part P_{ij} has 5.

Proposition (Vector representation of antisymmetric tensors). Any antisymmetric rank-2 tensor A_{ij} in $\mathbb{R}^3$ can be written as

$$A_{ij} = \varepsilon_{ijk}B_k, \quad \text{where} \quad B_k = \frac{1}{2}\varepsilon_{ijk}A_{ij}.$$

In matrix form,

$$(A_{ij}) = \begin{pmatrix} 0 & B_3 & -B_2 \\ -B_3 & 0 & B_1 \\ B_2 & -B_1 & 0 \end{pmatrix}.$$

Proof. Contracting $A_{ij} = \varepsilon_{ijk}B_k$ with $\varepsilon_{ij\ell}$ and using $\varepsilon_{ijk}\varepsilon_{ij\ell} = 2\delta_{k\ell}$ gives $\varepsilon_{ij\ell}A_{ij} = 2B_\ell$, hence $B_k = \frac{1}{2}\varepsilon_{ijk}A_{ij}$. Since ε_{ijk} is antisymmetric in i, j , we also have $B_k = \frac{1}{2}\varepsilon_{ijk}T_{ij}$, as the symmetric part of T_{ij} does not contribute. $\square$

Theorem (Full decomposition of a rank-2 tensor). Any rank-2 tensor T_{ij} in

$\mathbb{R}^3$ can be uniquely decomposed as

$$T_{ij} = P_{ij} + \varepsilon_{ijk}B_k + \frac{1}{3}\delta_{ij}Q,$$

where $Q = T_{kk}$ is the trace (1 component), $B_k = \frac{1}{2}\varepsilon_{pqk}T_{pq}$ is a vector (3 components), and $P_{ij} = \frac{1}{2}(T_{ij} + T_{ji}) - \frac{1}{3}\delta_{ij}Q$ is a symmetric traceless tensor (5 components).

Example (Strain rate tensor decomposition). Let $\mathbf{F}(\mathbf{r})$ be a vector field in $\mathbb{R}^3$. Its derivative $T_{ij} = \frac{\partial F_i}{\partial x_j}$ is a rank-2 tensor field. Applying the decomposition above gives the symmetric traceless piece

$$P_{ij} = \frac{1}{2} \left(\frac{\partial F_i}{\partial x_j} + \frac{\partial F_j}{\partial x_i} \right) - \frac{1}{3}\delta_{ij}\nabla \cdot \mathbf{F},$$

the antisymmetric piece $A_{ij} = \varepsilon_{ijk}B_k$, where

$$B_k = \frac{1}{2}\varepsilon_{ijk} \frac{\partial F_i}{\partial x_j} = -\frac{1}{2}(\nabla \times \mathbf{F})_k,$$

and the trace

$$Q = \frac{\partial F_k}{\partial x_k} = \nabla \cdot \mathbf{F}.$$

Hence a complete description of T_{ij} involves a scalar $\nabla \cdot \mathbf{F}$, a vector $\nabla \times \mathbf{F}$, and a symmetric traceless tensor P_{ij} .

14.2 The inertia tensor

Definition (Inertia tensor). Consider a system of point masses m_α at positions $\mathbf{r}_\alpha$ in $\mathbb{R}^3$. The *inertia tensor* of the system about the origin is the symmetric rank-2 tensor

$$I_{ij} = \sum_{\alpha} m_{\alpha} [|\mathbf{r}_{\alpha}|^2 \delta_{ij} - (\mathbf{r}_{\alpha})_i (\mathbf{r}_{\alpha})_j].$$

For a rigid body occupying a volume V with mass density $\rho(\mathbf{r})$, the sum is replaced by an integral:

$$I_{ij} = \int_V \rho(\mathbf{r}) (x_k x_k \delta_{ij} - x_i x_j) dV.$$

Proposition (Angular momentum of a rotating body). If the system rotates with angular velocity $\boldsymbol{\omega}$ about the origin, so that the velocity of mass m_α is $\mathbf{v}_\alpha = \boldsymbol{\omega} \times \mathbf{r}_\alpha$, then the total angular momentum satisfies

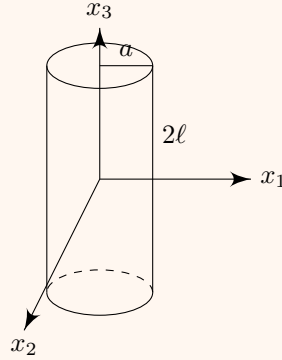
$$L_i = I_{ij}\omega_j.$$

Proof. The total angular momentum is

$$\begin{aligned}\mathbf{L} &= \sum_{\alpha} m_{\alpha} \mathbf{r}_{\alpha} \times \mathbf{v}_{\alpha} = \sum_{\alpha} m_{\alpha} \mathbf{r}_{\alpha} \times (\boldsymbol{\omega} \times \mathbf{r}_{\alpha}) \\ &= \sum_{\alpha} m_{\alpha} (|\mathbf{r}_{\alpha}|^2 \boldsymbol{\omega} - (\mathbf{r}_{\alpha} \cdot \boldsymbol{\omega}) \mathbf{r}_{\alpha}),\end{aligned}$$

using the vector identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{a}) = |\mathbf{a}|^2 \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{a}$. Writing in components gives $L_i = I_{ij} \omega_j$. $\square$

Example (Inertia tensor of a cylinder). Consider a cylinder of uniform density ρ_0 , radius a , and half-length ℓ , aligned along the x_3 -axis. The total mass is $M = 2\ell\pi a^2 \rho_0$.



Using cylindrical polar coordinates $x_1 = r \cos \theta$, $x_2 = r \sin \theta$, $x_3 = x_3$, with volume element $dV = r dr d\theta dx_3$, the diagonal components are:

$$I_{33} = \int_V \rho_0 (x_1^2 + x_2^2) dV = \rho_0 \int_0^a \int_0^{2\pi} \int_{-\ell}^{\ell} r^2 \cdot r dr d\theta dx_3 = \rho_0 \pi \ell a^4 = \frac{1}{2} M a^2.$$

Similarly,

$$\begin{aligned}I_{11} &= \int_V \rho_0 (x_2^2 + x_3^2) dV = \rho_0 \int_0^a \int_0^{2\pi} \int_{-\ell}^{\ell} (r^2 \sin^2 \theta + x_3^2) r dr d\theta dx_3 \\ &= \rho_0 \left(\pi a^2 \cdot \frac{2}{3} \ell^3 + 2\ell \int_0^a r^3 dr \int_0^{2\pi} \sin^2 \theta d\theta \right) = \rho_0 \pi a^2 \ell \left(\frac{a^2}{2} + \frac{2}{3} \ell^2 \right) = M \left(\frac{a^2}{4} + \frac{\ell^2}{3} \right),\end{aligned}$$

and by symmetry $I_{22} = I_{11}$.

For the off-diagonal elements, $I_{13} = -\int_V \rho_0 x_1 x_3 dV = 0$ since $\int_0^{2\pi} \cos \theta d\theta = 0$, and similarly all other off-diagonal elements vanish. Hence

$$I_{33} = \frac{1}{2} M a^2, \quad I_{11} = I_{22} = M \left(\frac{a^2}{4} + \frac{\ell^2}{3} \right).$$

In the particular case $\ell = \frac{a\sqrt{3}}{2}$, we have $I_{ij} = \frac{1}{2} M a^2 \delta_{ij}$, so that

$$\mathbf{L} = \frac{1}{2} M a^2 \boldsymbol{\omega}$$

for rotation about any axis.

14.3 Diagonalization of a symmetric second rank tensor

Remark. In matrix notation, the tensor transformation rule $T'_{ij} = R_{ip}R_{jq}T_{pq}$ becomes $T' = RTR^T = RTR^{-1}$.

Proposition (Diagonalization of symmetric tensors). A symmetric rank-2 tensor T_{ij} can be diagonalized by an orthogonal transformation. There exists an orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ of eigenvectors with real eigenvalues $\lambda_1, \lambda_2, \lambda_3$. The directions $\mathbf{e}_i$ are called the *principal axes*, and in Cartesian coordinates along these axes, $T_{ij} = \lambda_i \delta_{ij}$ (no sum).

Remark. For the inertia tensor, the eigenvalues are called the *principal moments of inertia*. The principal axes can often be identified from symmetries of the body, as in the cylinder example where the chosen axes yielded a diagonal tensor directly.

15 Invariant and isotropic tensors

15.1 Definitions and classification results

Definition (Invariant and isotropic tensor). A tensor T is *invariant* under a particular rotation R if

$$T'_{ij\dots k} = R_{ip}R_{jq}\cdots R_{kr}T_{pq\dots r} = T_{ij\dots k},$$

i.e. every component is unchanged under the rotation.

A tensor T which is invariant under every rotation is *isotropic*, i.e. the same in every direction.

Example (Isotropic tensors).

- (i) The Kronecker delta δ_{ij} is an isotropic rank-2 tensor.
- (ii) The Levi-Civita symbol ε_{ijk} is an isotropic rank-3 tensor.
- (iii) The inertia tensor of a uniform sphere about its centre is isotropic, by the rotational symmetry of the sphere.

Remark. The isotropy of δ_{ij} and ε_{ijk} ensures that the component definitions of the scalar and vector products, $\mathbf{a} \cdot \mathbf{b} = a_i b_i \delta_{ij}$ and $(\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk} a_j b_k$, are independent of the choice of Cartesian coordinate system.

Theorem (Classification of isotropic tensors in $\mathbb{R}^3$).

- (i) There are no isotropic tensors of rank 1, except the zero tensor.
- (ii) The most general rank 2 isotropic tensor is $T_{ij} = \alpha \delta_{ij}$ for some scalar α .
- (iii) The most general rank 3 isotropic tensor is $T_{ijk} = \beta \varepsilon_{ijk}$ for some scalar β .
- (iv) All isotropic tensors of higher rank are obtained by combining δ_{ij} and ε_{ijk} using tensor products, contractions, and linear combinations.

Proof (sketch). We analyze conditions for invariance under specific rotations through π or $\pi/2$ about coordinate axes.

- (i) Suppose T_i is rank-1 isotropic. Consider a rotation about x_3 through π :

$$(R_{ij}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Invariance requires $T_1 = R_{1p}T_p = R_{11}T_1 = -T_1$, hence $T_1 = 0$. Similarly, $T_2 = 0$. A rotation about x_1 gives $T_3 = 0$.

(ii) Suppose T_{ij} is rank-2 isotropic. Consider

$$(R_{ij}) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is a rotation through $\pi/2$ about the x_3 axis. Then

$$T_{13} = R_{1p}R_{3q}T_{pq} = R_{12}R_{33}T_{23} = T_{23}$$

and

$$T_{23} = R_{2p}R_{3q}T_{pq} = R_{21}R_{33}T_{13} = -T_{13}$$

So $T_{13} = T_{23} = 0$. Similarly, we have $T_{31} = T_{32} = 0$.

We also have

$$T_{11} = R_{1p}R_{1q}T_{pq} = R_{12}R_{12}T_{22} = T_{22}.$$

So $T_{11} = T_{22}$.

Rotations about other axes yield $T_{12} = T_{21}$ and $T_{22} = T_{33}$.

Hence $T_{ij} = \alpha\delta_{ij}$.

(iii) Suppose that T_{ijk} is rank-3 isotropic. Using the rotation by π about the x_3 axis, we have

$$T_{133} = R_{1p}R_{3q}R_{3r}T_{pqr} = -T_{133}.$$

So $T_{133} = 0$. We also have

$$T_{111} = R_{1p}R_{1q}R_{1r}T_{pqr} = -T_{111}.$$

So $T_{111} = 0$. We have similar results for π rotations about other axes and other choices of indices.

Then we can show that $T_{ijk} = 0$ unless all i, j, k are distinct.

Now consider

$$(R_{ij}) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

a rotation about x_3 through $\pi/2$. Then

$$T_{123} = R_{1p}R_{2q}R_{3r}T_{pqr} = R_{12}R_{21}R_{33}T_{213} = -T_{213}.$$

So $T_{123} = -T_{213}$. Along with similar results for other indices and axes of rotation, we find that T_{ijk} is totally antisymmetric, and $T_{ijk} = \beta\epsilon_{ijk}$ for some β . $\square$

Example (Rank 4 isotropic tensor). The most general isotropic tensor of rank 4 is

$$T_{ijkl} = \alpha\delta_{ij}\delta_{kl} + \beta\delta_{ik}\delta_{jl} + \gamma\delta_{il}\delta_{jk}$$

for some scalars α, β, γ .

Remark. Products involving ε_{ijk} reduce to expressions in δ_{ij} : for example, $\varepsilon_{ijp}\varepsilon_{klp} = \delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}$.

15.2 Application to invariant integrals

Theorem (Invariant integrals). Let

$$T_{ij\dots k} = \int_V f(\mathbf{x}) x_i x_j \cdots x_k \, dV.$$

where $f(\mathbf{x})$ is a scalar function and V is some volume.

Given a rotation R_{ij} , consider an *active* transformation: $\mathbf{x} = x_i \mathbf{e}_i$ is mapped to $\mathbf{x}' = x'_i \mathbf{e}_i$ with $x'_i = R_{ij}x_j$, i.e. we map the components but not the basis, and V is mapped to V' .

Suppose that under this active transformation,

- (i) $f(\mathbf{x}) = f(\mathbf{x}')$,
- (ii) $V' = V$ (e.g. if V is all of space or a sphere).

Then $T_{ij\dots k}$ is invariant under the rotation.

Proof. First note that the Jacobian of the transformation R is 1, since it is simply the determinant of R ($x'_i = R_{ip}x_p \Rightarrow \frac{\partial x'_i}{\partial x_p} = R_{ip}$), which is by definition 1. So $dV = dV'$.

Then we have

$$\begin{aligned} R_{ip}R_{jq} \cdots R_{kr} T_{pq\dots r} &= \int_V f(\mathbf{x}) x'_i x'_j \cdots x'_k \, dV \\ &= \int_V f(\mathbf{x}') x'_i x'_j \cdots x'_k \, dV \quad \text{using (i)} \\ &= \int_{V'} f(\mathbf{x}') x'_i x'_j \cdots x'_k \, dV' \quad \text{using (ii)} \\ &= \int_V f(\mathbf{x}) x_i x_j \cdots x_k \, dV \quad \text{since } x_i \text{ and } x'_i \text{ are dummy} \\ &= T_{ij\dots k}. \quad \square \end{aligned}$$

Remark. If conditions (i) and (ii) hold for *every* rotation R , then $T_{ij\dots k}$ is isotropic, which severely restricts its form.

Example (Coordinate integral over a sphere). Let V be the solid sphere $|\mathbf{r}| < a$ and consider

$$T_{ij} = \int_V x_i x_j \, dV.$$

The integrand has $f = 1$, which is invariant under rotations, and the solid sphere is invariant under any rotation. By the preceding theorem, T_{ij} is isotropic. By the classification theorem, the most general rank-2

isotropic tensor is $\alpha\delta_{ij}$, so

$$T_{ij} = \alpha\delta_{ij}$$

for some scalar α . Taking the trace gives

$$T_{ii} = 3\alpha = \int_V x_i x_i \, dV = 4\pi \int_0^a r^2 \cdot r^2 \, dr = \frac{4}{5}\pi a^5,$$

so

$$T_{ij} = \frac{4}{15}\pi a^5 \delta_{ij}.$$

This provides a rigorous justification for the common assertion that $T_{ij} = 0$ for $i \neq j$ “by symmetry.”

Example (Inertia tensor of a solid sphere). Consider a solid sphere of radius a , constant density ρ_0 , and total mass $M = \frac{4}{3}\pi a^3 \rho_0$. The inertia tensor is

$$I_{ij} = \int_V \rho_0 (x_k x_k \delta_{ij} - x_i x_j) \, dV.$$

Since $\int_V x_i x_j \, dV$ is isotropic (by the previous example) and $x_k x_k \delta_{ij}$ is manifestly isotropic, I_{ij} is isotropic. Using the result $T_{ij} = \frac{4}{15}\pi a^5 \delta_{ij}$ from above,

$$\begin{aligned} I_{ij} &= \rho_0 \left(\delta_{ij} \int_V x_k x_k \, dV - \int_V x_i x_j \, dV \right) = \rho_0 (\delta_{ij} T_{kk} - T_{ij}) \\ &= \rho_0 \left(\frac{4}{5}\pi a^5 \delta_{ij} - \frac{4}{15}\pi a^5 \delta_{ij} \right) = \frac{8}{15}\rho_0 \pi a^5 \delta_{ij} = \frac{2}{5} M a^2 \delta_{ij}. \end{aligned}$$